A CENTRALIZED SYSTEM FOR REAL-TIME FARM ANIMAL DISEASE REPORTING

INDIVIDUAL THESIS

Kumarasinghe S.S

IT22221414

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DECLARATION AND SUPERVISOR CERTIFICATION

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Student name: Kumarasinghe S.S

Student ID: IT22221414

Signature: _____ Date: _____

Supervisor certification. The above candidate carried out this research for the bachelor's degree thesis under my supervision.

Role and name	Signature	Date
Supervisor: Mrs. Dilani Lunugalage		
Co-supervisor: Mr. Eishan Weerasinghe		

ABSTRACT

Livestock disease outbreaks remain a concern for animal health, agricultural productivity, and rural livelihoods in Sri Lanka. Foot-and-Mouth Disease (FMD) and Lumpy Skin Disease (LSD) are difficult to model using conventional approaches because the available veterinary surveillance records are sparse, irregular, and highly imbalanced. This research developed a climate-informed spatio-temporal decision-support component for district-level FMD and LSD outbreak-risk estimation. When climatic information for the assessed month is available, the model operates as a concurrent district-month nowcasting classifier by combining current climatic conditions with recent epidemiological and environmental history. The software also supports future-date risk projection; however, these projections use the most recent historical conditions from the same calendar month as a climatological proxy and are therefore not treated as independently validated multi-month forecasts.

The research reconstructed veterinary surveillance records into structured district-month datasets and combined them with climatic, seasonal, livestock-density, teleconnection, spatial-neighbor, and historical outbreak features. The FMD dataset contained 2,400 district-month observations from 2017 to 2024, while the LSD dataset contained 1,500 observations from 2020 to 2024. The models were assessed using time-aware and year-separated evaluation procedures, with particular attention to class imbalance and leakage control. Mondrian conformal prediction was evaluated offline as an uncertainty-quantification method for FMD, while runtime explanations were generated through exact linear log-odds decomposition of the fitted models.

For FMD, the 30-feature Logistic Regression model served as the production baseline, while an enhanced 31-feature variant added previous-month district outbreak status through own_outbreak_lag1. The enhanced model achieved a mean walk-forward ROC-AUC of 0.8120 across the 2022–2024 evaluation folds and a ROC-AUC of 0.8166 with PR-AUC of 0.6968 on the held-out 2024 fold. At a decision threshold of 0.40, it detected 67.9% of the observed 2024 outbreaks. For LSD, the deployed 28-feature Platt-calibrated ElasticNet model achieved a five-fold Leave-One-Year-Out mean ROC-AUC of 0.6136 and mean PR-AUC of 0.2699 across 2020–2024. These results indicate useful predictive signal, although year-to-year variation and limited positive observations require cautious interpretation.

The principal contribution of this individual research is the development of an interpretable and uncertainty-aware disease-risk component that converts sparse surveillance data into district-level advisory information for veterinary decision-making. The system is intended to support risk prioritization and professional veterinary judgement rather than replace diagnosis or official disease-control decisions.

Keywords: Foot-and-Mouth Disease, Lumpy Skin Disease, Disease Risk Nowcasting, Spatio-Temporal Machine Learning, Veterinary Decision Support

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LIST OF ABBREVIATIONS

Table 1 : List of Abbreviations

Abbreviation	Meaning
API	Application Programming Interface
AUC	Area Under the Curve
CHIRPS	Climate Hazards Group InfraRed Precipitation with Station Data
CI	Confidence Interval
CV	Cross-Validation
DAPH	Department of Animal Production and Health
DMI	Dipole Mode Index
ENSO	El Niño–Southern Oscillation
ERA5	ECMWF Reanalysis Version 5
FMD	Foot-and-Mouth Disease
GADM	Database of Global Administrative Areas
GLW	Gridded Livestock of the World
IOD	Indian Ocean Dipole
LOYO	Leave-One-Year-Out
LR	Logistic Regression
LSD	Lumpy Skin Disease
ML	Machine Learning
NINO3.4	Niño 3.4 Sea-Surface Temperature Index
OOF	Out-of-Fold
PR-AUC	Precision–Recall Area Under the Curve
RBAC	Role-Based Access Control
ROC-AUC	Receiver Operating Characteristic Area Under the Curve
SDG	Sustainable Development Goal
UI	User Interface
XAI	Explainable Artificial Intelligence

1 INTRODUCTION

1.1 Overall project context

Livestock production contributes to food security, rural livelihoods, and agricultural activity in Sri Lanka. Disease outbreaks affecting cattle and buffalo can reduce animal productivity, increase veterinary costs, disrupt livestock movement, and place additional pressure on farmers and animal-health services. Foot-and-Mouth Disease (FMD) and Lumpy Skin Disease (LSD) are two important livestock diseases that require continued surveillance because their occurrence varies across locations and time and can be influenced by epidemiological and environmental conditions.

Traditional disease-surveillance approaches generally depend on identifying and reporting cases after disease occurrence. Statistical surveillance methods such as the Farrington algorithm and the Early Aberration Reporting System have been used to detect unusual disease activity from historical records [1], [2]. However, such methods rely on sufficiently regular historical observations, whereas livestock surveillance records may contain irregular reporting periods and long intervals with no recorded outbreaks [3]. This creates difficulties when conventional time-series or machine-learning approaches are applied directly to veterinary data.

In Sri Lanka, historical disease information is available through veterinary epidemiology reports published by the Department of Animal Production and Health (DAPH). These reports provide useful disease information, but they do not always form a complete and regular district-level monthly time series. One requirement of this study was therefore to restructure available surveillance records into a consistent district-month representation suitable for computational analysis.

This study was conducted as the disease-risk component of the wider **R26-SE-013: A Centralized System for Real-Time Farm Animal Disease Reporting** project. The broader system combines several livestock-health functions within one digital platform, while this thesis focuses specifically on climate-informed disease-risk modelling.

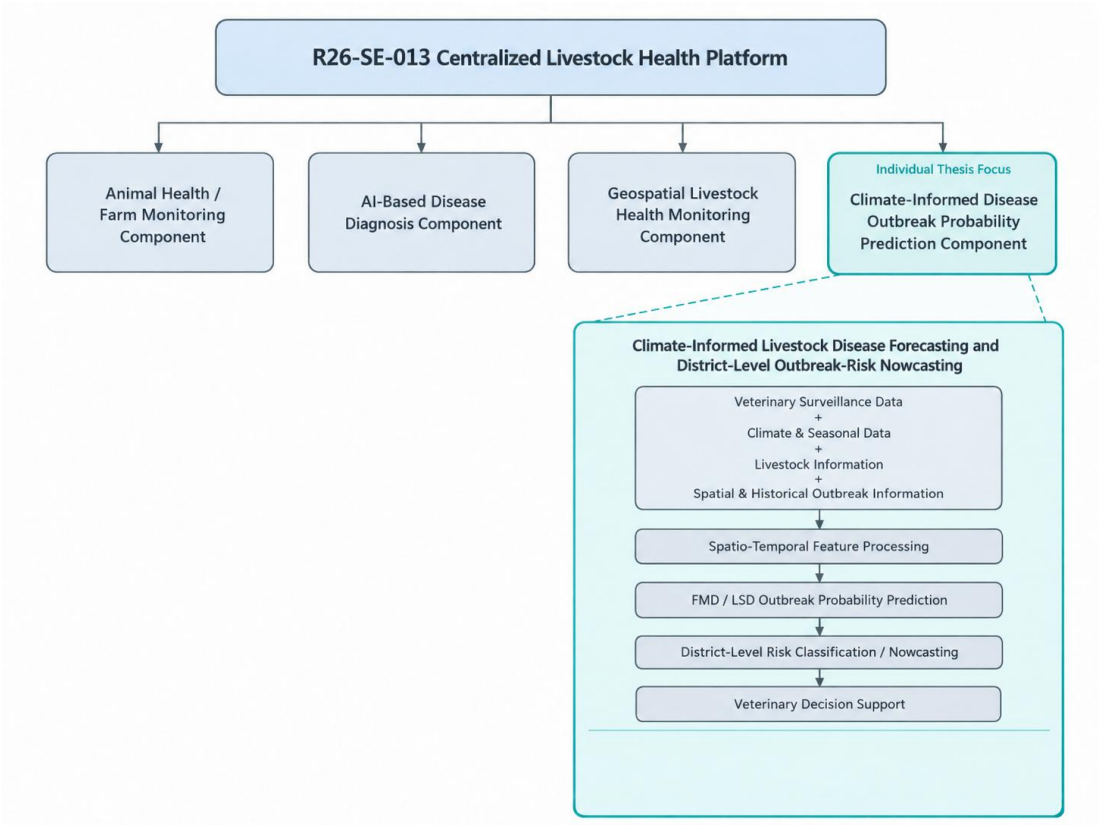


Figure 1 : Position of the Disease-Risk Component within the R26-SE-013 Livestock Health Platform

The component was initially developed under the broader description of disease forecasting. Examination of the final implementation showed that two related forms of risk estimation are supported. When climatic observations for the requested district-month are available, the model performs concurrent outbreak-risk nowcasting using information from that month together with historical lag features. For future dates where observed climatic values are unavailable, the software generates a climatological risk projection using the latest available historical conditions from the same calendar month. The latter should be interpreted as scenario-based risk estimation rather than validated multi-month disease forecasting.

The study covers all 25 administrative districts of Sri Lanka and focuses on FMD and LSD. The FMD data span 2017–2024, while the LSD data span 2020–2024. The overall aim is to convert sparse veterinary surveillance and related environmental information into district-level risk estimates that can support veterinary monitoring and prioritisation.

1.2 Individual component context

The disease-risk component developed in this study estimates district-level outbreak risk for FMD and LSD from sparse veterinary surveillance and related environmental data.

The main analytical unit is the **district-month**. This representation allows disease conditions to be examined across both space and time. Differences between districts may be associated with monsoon patterns, livestock distribution, environmental conditions, previous outbreak activity, and disease occurrence in neighbouring districts.

The reconstructed FMD dataset contains **2,400 district-month observations from 2017 to 2024**, while the LSD dataset contains **1,500 observations from 2020 to 2024**. The datasets are highly imbalanced, with outbreak-positive observations representing only a small proportion of all district-months. This means that overall accuracy alone is not sufficient for evaluating model usefulness, because a classifier could favour the majority non-outbreak class while still missing important outbreak events.

The modelling framework therefore combines several categories of information, including climatic observations, seasonal and monsoon indicators, livestock-density variables, large-scale climate indices, neighbouring outbreak history, and recent outbreak activity within the same district.

The modelling architecture contains two stages. **Stage 1** estimates whether an outbreak is likely to occur within a district-month. **Stage 2** is designed to provide a conditional estimate of outbreak severity when elevated outbreak risk is identified. The primary research emphasis of this thesis is Stage 1 outbreak-risk classification because it has the stronger and more complete evaluation evidence.

For FMD, the production pipeline uses a 30-feature Logistic Regression model as the default baseline. An enhanced 31-feature variant additionally includes own_outbreak_lag1, representing reported outbreak activity in the same district during the previous month, when that information is available and verified. For LSD, the primary production model is a calibrated 28-feature ElasticNet classifier, with a 27-feature fallback model used when previous-month outbreak information is unavailable.

The component also provides information beyond a raw probability estimate. Runtime predictions apply an operational decision threshold, provide risk and severity information where applicable, report data-quality status, and expose feature-level explanations through exact linear log-odds decomposition. During the research evaluation, Mondrian conformal prediction was additionally examined as an uncertainty-quantification approach for FMD. It is therefore treated in this thesis as a validated research method rather than as a fully deployed runtime confidence mechanism.

1.3 Problem statement

Livestock disease-risk modelling in Sri Lanka is affected by several connected problems.

The first problem is the structure of the available surveillance data. FMD and LSD outbreaks occur less frequently than non-outbreak periods, and the reconstructed datasets therefore contain relatively few positive district-month observations. This class imbalance limits the number of outbreak examples available for model development and means that overall accuracy alone is not an appropriate measure of model usefulness.

A second problem is that disease risk cannot be represented adequately using historical case information alone. Environmental and climatic conditions can influence pathogen survival, vector activity, and disease transmission. Previous livestock-disease studies have reported the importance of environmental and spatial factors, including climatic conditions, livestock distribution, and geospatial characteristics associated with FMD and LSD occurrence [4] – [9].

A third problem concerns evaluation. Disease data are temporally ordered, and randomly mixing observations from different years can allow information from later periods to influence earlier test cases. This can produce optimistic performance estimates. For this reason, the models in this study require time-aware and year-separated evaluation procedures that better reflect the structure of the available disease data.

Class imbalance introduces another difficulty. From a veterinary perspective, detecting outbreak periods is more important than achieving a high overall accuracy score. Model evaluation therefore needs to consider recall, ROC-AUC, PR-AUC, and the effect of classification-threshold selection.

Finally, a disease-risk estimate intended for professional use should provide more information than a single probability. Users need some understanding of the factors influencing a prediction and of the uncertainty surrounding model performance. In this study, model interpretability was incorporated directly into the operational prediction process, while formal uncertainty quantification was investigated separately during model evaluation.

The research problem addressed in this thesis is:

How can sparse and imbalanced livestock disease surveillance data be combined with climatic, seasonal, spatial, livestock-related, and historical epidemiological information to estimate interpretable and uncertainty-aware district-level FMD and LSD outbreak risk under temporally valid evaluation conditions?

1.4 Research gap

Existing disease-surveillance and prediction research provides useful methods, but several limitations remain when these approaches are applied to Sri Lankan livestock surveillance.

Traditional outbreak-detection methods such as the Farrington algorithm and EARS identify unusual disease activity using historical surveillance data [1], [2]. These approaches are useful when data are sufficiently regular, but their performance can be affected when disease records are sparse or irregular [3].

Statistical count-based approaches can be useful when disease records contain many periods without reported outbreaks. However, these approaches generally focus on the structure of disease occurrence or count and do not by themselves address the wider combination of climatic, spatial, livestock-related, and recent epidemiological information required in this study.

More recent machine-learning research has included meteorological and geospatial variables in livestock disease prediction. Studies on LSD have demonstrated the potential value of environmental predictors and ensemble learning methods [8], [9].

Despite these developments, there is still a need for an approach that combines the following within one Sri Lankan district-level framework: reconstruction of sparse surveillance records, class-imbalance handling, climate and seasonality, livestock-related features, spatial and temporal outbreak history, time-aware and year-separated validation, prediction uncertainty, and interpretable outputs.

A further gap concerns temporal evaluation. Strong results obtained using random or in-sample testing do not necessarily demonstrate that a model will generalise to later years. For a disease-risk model intended to support real-world monitoring, time-aware evaluation is therefore important.

The research gap addressed in this thesis is the limited availability of a Sri Lankan district-level livestock disease-risk approach that combines sparse-data reconstruction, climate-informed spatio-temporal modelling, leakage-aware temporal validation, uncertainty estimation, and interpretable risk outputs for both FMD and LSD.

This study addresses the gap through the development and evaluation of a climate-informed spatio-temporal disease-risk component that supports concurrent district-month nowcasting and climatological risk projection for veterinary decision support.

1.5 Aim and objectives

1.5.1 Aim

The aim of this study is:

To develop and evaluate a climate-informed spatio-temporal approach for estimating district-level FMD and LSD outbreak risk in Sri Lanka from sparse veterinary surveillance and related environmental information, while providing interpretable outputs and evaluating uncertainty-aware methods for veterinary decision support.

1.5.2 Research objectives

The research objectives are:

1. To reconstruct sparse FMD and LSD surveillance information into consistent district-month datasets covering the 25 administrative districts of Sri Lanka.
2. To integrate climatic, seasonal, livestock-related, teleconnection, spatial-neighbour, and historical outbreak information into suitable spatio-temporal feature sets.
3. To develop and evaluate calibrated machine-learning models for district-month outbreak-risk classification under highly imbalanced disease data.
4. To examine whether recent district-level outbreak autocorrelation improves FMD outbreak-risk discrimination compared with the baseline feature representation.
5. To evaluate model performance using time-aware and year-separated validation procedures, together with leakage-control measures appropriate to the available disease datasets.
6. To investigate classification thresholds that provide an appropriate balance between outbreak detection and false-positive alerts.
7. To evaluate Mondrian conformal prediction as an uncertainty-quantification method and to provide interpretable information about the factors influencing individual risk estimates.

The software-engineering objective of the study is:

To integrate the developed disease-risk models into an API-based decision-support component and user interface within the wider R26-SE-013 livestock-health platform.

1.6 Research questions or hypotheses

The study is guided by the following research questions:

RQ1: How can sparse and irregular FMD and LSD surveillance records be transformed into consistent district-month datasets suitable for machine-learning analysis?

RQ2: To what extent can climatic, seasonal, livestock-related, spatial, and historical outbreak features discriminate between outbreak and non-outbreak district-months under time-aware or year-separated evaluation?

RQ3: Does the inclusion of recent district-level outbreak autocorrelation improve FMD outbreak-risk discrimination compared with the baseline model?

RQ4: How consistently does the selected LSD model perform across different held-out years under Leave-One-Year-Out evaluation?

RQ5: How does the selected classification threshold affect the balance between outbreak recall and false-positive alerts?

RQ6: Can probability calibration, offline conformal uncertainty evaluation, and interpretable feature attribution provide useful supporting information alongside district-level outbreak-risk estimates?

A separate product-oriented question is considered during implementation:

Table 2 : Mapping of Research Questions to Objectives

Research Question	**Related Objective(s)**
RQ1 – Reconstruction of sparse surveillance data	Objective 1
RQ2 – Value of spatio-temporal features	Objectives 2 and 3
RQ3 – Effect of outbreak autocorrelation	Objective 4
RQ4 – LSD temporal generalization	Objective 5
RQ5 – Threshold trade-off	Objective 6
RQ6 – Uncertainty and interpretability	Objective 7
PQ1 – Software integration	Engineering Objective

1.7 Scope limitations and assumptions

1.7.1 Scope

The study is geographically limited to the **25 administrative districts of Sri Lanka** and considers two livestock diseases:

- Foot-and-Mouth Disease;
- Lumpy Skin Disease.

The temporal coverage is:

- **FMD:** 2017–2024;
- **LSD:** 2020–2024.

The primary modelling task is district-month outbreak-risk classification. For historical or currently observed periods where target-month climatic information is available, the model operates as a concurrent nowcasting system. For future requested dates, the implemented forecast endpoints use the latest historical observations from the same calendar month as climatological proxies. These future outputs are therefore interpreted as climatological risk projections rather than independently validated future-weather forecasts.

The work includes data reconstruction, feature engineering, calibrated classification, temporal and spatial lag modelling, time-aware and year-separated evaluation, threshold analysis, uncertainty estimation, model interpretation, backend inference, and interface integration.

The main FMD production baseline is a 30-feature Logistic Regression model. A 31-feature autocorrelation variant is also evaluated and may be used when verified previous-month outbreak status is available. The primary LSD production model is a calibrated 28-feature ElasticNet classifier, with a dedicated 27-feature fallback model for cases where recent outbreak history is unavailable.

1.7.2 Limitations

Sparse positive observations - Outbreak months form a small proportion of both datasets, which limits the number of positive examples available for model training and evaluation.

Year-to-year variation in LSD performance - The deployed LSD model achieved a five-fold LOYO mean ROC-AUC of 0.6136 and PR-AUC of 0.2699, but performance varied between individual held-out years. The limited five-year observation period and small number of outbreak-positive district-months therefore restrict strong claims about stable temporal generalization.

Reliance on recent outbreak history - Some final features represent previous outbreak activity. These variables can improve temporal representation, but the resulting relationships should be interpreted as predictive associations rather than evidence of causation.

Data provenance - Formal independent certification of some DAPH-derived records was not available during this study. This limitation should be considered when interpreting the dataset provenance.

Imputation procedure - Fallback missing-value handling used static multi-year median values in part of the pipeline. This introduces a potential leakage risk. A fully time-bounded expanding-window imputation procedure would provide a stricter future implementation.

Prediction-horizon limitation - The main models require several concurrent target-month climatic variables. Future-date requests therefore rely on historical same-calendar-month proxy conditions rather than future meteorological forecasts. The system can produce future-period climatological risk estimates, but these should not be interpreted as validated one-to-three-month-ahead disease forecasts.

Decision-support boundary - The system does not diagnose individual animals or determine veterinary interventions automatically. Its outputs are intended to support professional monitoring and prioritisation.

1.7.3 Assumptions

The study assumes that the reconstructed surveillance records provide a reasonable representation of reported district-level disease occurrence during the study period.

It also assumes that environmental variables assigned to each district-month reasonably represent the conditions associated with that location and period.

Spatial-neighbour features use administrative adjacency as a representation of local disease context. Actual disease transmission may also depend on livestock movement, road networks, farm-level contact, and other factors not fully represented by district boundaries.

The interpretation of model outputs also assumes that future operating conditions remain reasonably similar to the historical data used during development. Major

changes in surveillance practices, livestock movement, disease epidemiology, or climatic patterns may affect model performance.

1.8 Individual contribution

This thesis presents my individual contribution to the R26-SE-013 project through the development and evaluation of the climate-informed disease outbreak-risk component.

My research contribution includes the reconstruction of sparse disease-surveillance data into district-month datasets, development of the feature pipeline, and integration of climatic, seasonal, livestock, teleconnection, spatial-neighbour, and historical outbreak information.

The modelling work included the development and evaluation of FMD and LSD risk classifiers, analysis of recent outbreak autocorrelation, time-aware and year-separated model evaluation, class-imbalance assessment, threshold analysis, uncertainty estimation, and interpretable prediction outputs.

A key methodological contribution was the use of evaluation procedures that preserved the temporal structure of the available data rather than relying only on random train-test splitting. FMD was assessed using expanding-window walk-forward evaluation, including a held-out 2024 test fold, while LSD was assessed using year-separated Leave-One-Year-Out evaluation. This allowed performance differences across time periods to remain visible rather than being hidden within a single full-dataset score.

The product contribution involved integrating the disease-risk models into the wider application through FastAPI services and a React-based interface. Separate FMD and LSD prediction endpoints are exposed under /api/v1/risk-forecasting/predict/, while additional forecast endpoints generate 25-district climatological risk projections. The interface presents risk probabilities, conditional severity information, explanation outputs, and role-appropriate decision-support information.

The individual contribution therefore covers the complete path from data preparation and feature engineering to model development, statistical evaluation, interpretability and uncertainty research, and integration into the final software system.

1.9 Thesis structure

This thesis is organised into nine chapters.

Chapter 1 – Introduction presents the project context, disease-risk component, research problem, research gap, aim, objectives, research questions, scope, limitations, assumptions, and individual contribution.

Chapter 2 – Literature Review and Related Work reviews livestock disease surveillance, FMD and LSD modelling, sparse and zero-heavy epidemiological data, environmental and climatic disease drivers, machine-learning approaches, spatio-temporal modelling, uncertainty estimation, and model interpretability. It concludes with a comparison of existing work and refinement of the research gap.

Chapter 3 – Research Methodology describes the research design, data sources, district-month reconstruction process, preprocessing, feature engineering, model development, time-aware and year-separated validation, evaluation metrics, uncertainty methods, and reproducibility procedures.

Chapter 4 – Product and Solution Design presents the functional and non-functional requirements, component architecture, design decisions, prediction workflow, and the relationship between the disease-risk component and the wider system.

Chapter 5 – Implementation, Integration and Testing explain the development environment, model implementation, backend API, frontend integration, automated testing, and operational behaviour of the component.

Chapter 6 – Results and Validation presents the FMD and LSD modelling results, time-aware and year-separated evaluation, threshold analysis, uncertainty-quantification results, model interpretation, software-testing outcomes, and component-level validation.

Chapter 7 – Discussion interprets the results in relation to the research questions and previous studies. It discusses differences between FMD and LSD performance, feature contributions, temporal generalisation, limitations, and practical implications.

Chapter 8 – Commercialisation and Sustainability considers the potential value of the component to veterinary and agricultural stakeholders, deployment and sustainability options, scalability, intellectual-property considerations, compliance issues, and links to relevant Sustainable Development Goals.

Chapter 9 – Conclusion and Future Directions summarises the main findings, evaluates the achievement of the research objectives, presents the individual research and product contributions, and identifies directions for further development.

The thesis concludes with the reference list, glossary where required, and appendices containing supplementary research and technical material.

2 LITERATURE REVIEW AND RELATED WORK

2.1 Review method and scope

This chapter reviews research relevant to the development of a climate-informed livestock disease-risk nowcasting component for Foot-and-Mouth Disease (FMD) and Lumpy Skin Disease (LSD). The review was conducted as a targeted narrative review rather than a formal systematic review. Its purpose was to identify the concepts, methods, and limitations most directly related to the research problem addressed in this thesis.

The literature was examined across several connected areas. These included conventional outbreak-surveillance methods, the epidemiological characteristics of FMD and LSD, climatic and spatial factors associated with disease occurrence, modelling of sparse and imbalanced disease data, machine-learning approaches for livestock disease prediction, temporal validation, probability calibration, uncertainty quantification, and model interpretability.

Priority was given to peer-reviewed journal articles, established methodological publications, and authoritative veterinary or environmental sources. Attention was given to research published on spatial and spatio-temporal livestock disease modelling because the present study represents disease occurrence at district-month level rather than as an independent national time series.

The review also distinguishes between outbreak detection, forecasting, nowcasting, and climatological risk projection. Outbreak-detection methods identify abnormal activity from surveillance data, while future forecasting estimates a later outcome using only information available before the target period. When target-month climatic observations are available, the implemented models in this study are more accurately described as district-month outbreak-risk nowcasting models. For future-date requests, the software instead uses historical same-calendar-month environmental conditions as proxies and therefore produces climatological risk projections rather than validated future-weather forecasts.

The literature review was used for two purposes. First, it provided the theoretical basis for selecting climatic, temporal, spatial, and epidemiological features. Second, it helped identify methodological weaknesses in existing approaches that were relevant to the design of the present study, particularly sparse surveillance data, class imbalance, temporal leakage, uncertainty, and interpretability.

2.2 Domain and theoretical foundations

2.2.1 Livestock Disease Surveillance and Outbreak Detection

Disease surveillance aims to identify changes in disease occurrence early enough to support appropriate investigation and control. Traditional statistical surveillance has commonly relied on historical observations to determine whether the number of reported cases during a particular period differs substantially from an expected baseline.

Farrington et al. developed a statistical algorithm for the early detection of infectious-disease outbreaks using historical surveillance counts [1]. The Early Aberration Reporting System later provided another approach for detecting unusual patterns from routinely collected surveillance information [2]. A broader review by Buckeridge et al. showed that outbreak-detection algorithms differ substantially in their assumptions, sensitivity, and behaviour under different disease-reporting conditions [3]. These methods established an important foundation for automated surveillance, but they generally rely on the availability of reasonably consistent historical data.

This becomes more difficult for livestock diseases where records may contain irregular reporting, long periods without recorded outbreaks, and differences in reporting quality between locations. In such circumstances, the data cannot always be treated as a conventional continuous disease-count time series. The present research therefore uses a reconstructed district-month representation, allowing reported disease occurrence to be aligned with climate and other explanatory variables.

2.2.2 Foot-and-Mouth Disease

FMD is a highly contagious viral disease affecting cloven-hoofed livestock and remains an important transboundary animal disease in many parts of Africa and Asia.

Its epidemiology is influenced by animal movement, livestock density, immunity, virus characteristics, management practices, and environmental conditions.

A recent scoping review by González Gordon et al. examined spatial and spatio-temporal studies of FMD outbreaks in endemic settings and found that FMD risk is commonly associated with livestock distribution, movement, geographical proximity, seasonality, and environmental conditions [4]. The review also reported considerable variation in seasonal patterns between locations, indicating that climatic relationships should not be assumed to be identical across different countries.

Environmental persistence is another relevant consideration. Mielke and Garabed examined experimental evidence on the survival of FMD virus under different environmental conditions and found that temperature alone was insufficient to describe virus survival. Interactions between temperature and relative humidity affected environmental persistence, with different survival patterns under wet and dry conditions [5].

These findings support the inclusion of climatic and seasonal information in FMD risk modelling. However, they also suggest that environmental variables should be interpreted as risk indicators rather than direct causal explanations of an outbreak. Disease transmission remains dependent on interacting biological, livestock, spatial, and management factors.

2.2.3 Lumpy Skin Disease

LSD is a viral disease affecting bovines and is strongly associated with arthropod-mediated transmission. This makes environmental and climatic conditions especially relevant because vector activity can change according to temperature, rainfall, humidity, and habitat conditions.

Spatial studies have repeatedly shown that LSD occurrence is geographically structured. Allepuz et al. analysed LSD occurrence across Eurasia using outbreak records together with climatic, land-cover, and cattle-density information. Their study demonstrated the value of spatial risk modelling for identifying areas with conditions favourable for disease occurrence [6].

Afshari Safavi evaluated several machine-learning methods for predicting LSD occurrence using meteorological, geospatial, livestock-density, land-cover, and elevation variables. The study found that machine-learning models could discriminate between affected and unaffected observations and reported strong predictive performance for some model configurations [7]. However, results obtained in one geographical setting cannot automatically be transferred to another country because surveillance quality, disease prevalence, livestock structure, and environmental relationships may differ.

More recent ecological niche studies have further demonstrated the spatial nature of LSD risk. Li et al. modelled the spatio-temporal distribution of LSD in Asia and identified cattle density, land cover, buffalo density, and temperature-related variables among the important contributors to predicted suitability [8]. An et al. similarly used ecological niche modelling for global LSD risk assessment and identified livestock-density and bioclimatic factors as important predictors [9].

Together, these studies support the use of environmental, livestock, and spatial information for LSD risk modelling. They also show that LSD risk is unlikely to be explained by a single environmental variable.

2.2.4 Climate, Seasonality and Large-Scale Climate Variability

Climate influences disease systems both directly and indirectly. Temperature and humidity may affect pathogen survival, while rainfall and seasonal conditions can influence vegetation, animal movement, water availability, vector populations, and livestock contact patterns.

For Sri Lanka, rainfall is strongly seasonal and is influenced by monsoonal circulation as well as large-scale ocean-atmosphere processes. Zubair demonstrated that the Indian Ocean Dipole can substantially modulate rainfall during the Sri Lankan Maha season [10]. Later work on ENSO also demonstrated that El Niño-Southern Oscillation conditions can influence rainfall differently according to season and region within Sri Lanka [11].

These climatic relationships provide a theoretical basis for including large-scale climate indices such as Niño 3.4 and the Indian Ocean Dipole Mode Index as

contextual variables. They are not treated as direct disease causes. Instead, they represent broader climatic variability that may influence local environmental conditions associated with livestock disease risk.

Seasonality is also represented directly in the present research through month-based cyclic variables and monsoon indicators. This allows the model to distinguish recurring annual patterns without treating December and January as unrelated endpoints of a linear calendar variable.

2.2.5 Spatial and Temporal Dependence

Disease occurrence in one district may be related to events in nearby districts. Geographic proximity can reflect livestock movement, shared environmental conditions, trade relationships, vector movement, or other mechanisms that are not visible when districts are analysed independently.

Spatial disease studies for both FMD and LSD have demonstrated geographical clustering and non-random distribution of outbreaks [4], [6], [8]. This supports the use of neighbouring outbreak information as part of a district-level feature representation.

Temporal dependence is equally important. An outbreak in a particular district may increase the probability of continued or recurrent disease activity during the following period. For this reason, lagged outbreak information can provide useful predictive information.

However, temporal and spatial dependence also create a methodological problem during evaluation. If observations that are closely related in time or space are randomly distributed between training and testing sets, model performance may be overestimated. Roberts et al. showed that ignoring spatial or temporal structure during cross-validation can produce overly optimistic estimates of predictive performance [12].

This provides the theoretical justification for the time-aware and year-separated validation procedures adopted in this thesis.

2.2.6 Class Imbalance and Evaluation Metrics

Livestock outbreak datasets are naturally imbalanced because outbreaks occur less frequently than non-outbreak periods. A classifier can therefore achieve high accuracy by repeatedly predicting the majority class while identifying relatively few true outbreaks.

For this reason, accuracy alone is not an appropriate primary evaluation measure. Sensitivity or recall provides information about the proportion of true outbreaks identified, while precision indicates how many predicted outbreak alerts correspond to observed outbreaks.

ROC-AUC is useful for measuring ranking ability across decision thresholds, but it may provide an overly favourable impression when the negative class substantially outnumbers the positive class. Saito and Rehmsmeier demonstrated that precision-recall analysis is particularly informative for imbalanced binary classification because it directly reflects performance on the positive class [13].

This literature supports the use of both ROC-AUC and PR-AUC in the present study, together with threshold-specific recall and false-positive analysis.

2.2.7 Probability Calibration

A machine-learning classifier may rank observations correctly without producing probabilities that correspond closely to observed outcome frequencies. This matters in a risk-based application because a probability such as 0.70 should ideally represent a meaningfully higher level of risk than a probability such as 0.30.

Niculescu-Mizil and Caruana compared probability estimates generated by several supervised learning algorithms and showed that calibration methods such as Platt scaling and isotonic regression can improve probability estimates when the underlying classifier produces systematically distorted scores [14].

Probability calibration is therefore relevant to the present study because the model output is communicated as risk information rather than only as a binary class.

2.2.8 Temporal Validation

Conventional random cross-validation assumes that observations are sufficiently independent and exchangeable. That assumption becomes problematic when observations are ordered in time.

Cerqueira et al. compared several performance-estimation methods for time-series problems and showed that evaluation procedures can produce meaningfully different estimates when temporal dependency and non-stationarity are present. Their findings support the use of out-of-sample and time-aware evaluation when the objective is to estimate performance on future or later observations [15].

Although the present research concerns district-month classification rather than conventional univariate time-series forecasting, the same principle applies. A model intended to operate on later months or years should be evaluated using earlier periods for training and later periods for testing wherever possible.

2.2.9 Uncertainty Quantification

A point probability does not communicate all uncertainty associated with a prediction. This is particularly important when disease records are sparse and model performance changes between years.

Conformal prediction provides a model-agnostic framework for producing prediction sets with statistical coverage properties under defined assumptions. Vovk, Gammerman and Shafer provided the foundational treatment of conformal prediction [16]. More recently, Angelopoulos and Bates presented a practical overview of conformal methods and their use for distribution-free uncertainty quantification [17].

Class-conditional, or Mondrian, conformal prediction is particularly relevant when classes are imbalanced because calibration can be performed separately according to class. In this research, Mondrian conformal prediction was evaluated offline as an additional uncertainty-quantification method for the FMD classifier. Its performance is therefore reported as part of the research evaluation rather than presented as a fully deployed runtime confidence mechanism.

2.2.10 Interpretability

Interpretability is important when machine-learning outputs are used to support decisions in domains where the consequences of incorrect predictions may be significant.

Rudin argued that interpretable models should be preferred where possible in high-stakes applications rather than relying entirely on post-hoc explanations of complex black-box models [18].

This position is relevant to the final model design of the present study. Logistic models provide a direct mathematical relationship between predictors and log-odds. For a linear model, the contribution of an individual feature can therefore be calculated directly from the fitted coefficient and the feature value relative to its reference level. This avoids presenting an explanation that is disconnected from the actual decision function of the model.

SHAP was also used during offline exploratory analysis to examine global feature behavior, while the deployed prediction services use exact coefficient-based log-odds decomposition for individual predictions.

2.3 Related research

2.3.1 Statistical Outbreak Detection

Early outbreak-detection research focused primarily on identifying abnormal increases in surveillance counts. Farrington et al. [1] and Hutwagner et al. [2] are important examples of this approach. Such methods are useful when surveillance systems produce regular observations and the main objective is to identify deviations from a historical baseline.

Their limitation for the present research is that the available livestock disease information is not a clean, continuous count series for every district-month. In addition, these methods do not naturally combine multiple climatic, livestock, and spatial predictors.

2.3.2 Spatial and Spatio-Temporal FMD Research

The FMD literature has increasingly moved from purely temporal analysis toward spatial and spatio-temporal approaches. González Gordon et al. reviewed studies conducted in endemic settings and found that spatial clustering, seasonality, animal density, movement, and geographical relationships are recurring themes in FMD risk analysis [4].

An important observation from this literature is that FMD seasonality is not consistent across all locations. Some studies report peaks during dry periods, while others identify associations with rainy or cooler seasons. This suggests that environmental effects are context-dependent and supports the use of locally structured modelling rather than transferring a fixed climate-risk relationship from another country.

The present research follows this direction by combining district-specific observations with climate, seasonal indicators, neighbouring outbreak activity, and recent local outbreak history.

2.3.3 Machine Learning for LSD Risk Prediction

LSD research provides clearer examples of environmental and machine-learning risk modelling because arthropod-mediated transmission makes climatic conditions particularly relevant.

Afshari Safavi [7] compared machine-learning methods using meteorological and geospatial variables and reported high predictive performance for some configurations. The study demonstrates that non-linear machine-learning methods can identify relationships between environmental conditions and LSD occurrence.

However, a high-test score in one study does not guarantee temporal stability in another setting. The epidemiological context, prevalence, spatial coverage, validation method, and disease-reporting process can all influence measured performance. This is particularly relevant to the present study because its LSD model showed substantial variation between held-out years.

Ecological niche modelling studies provide another approach. Allepuz et al. [6], Li et al. [8], and An et al. [9] modelled spatial suitability or risk using combinations of

climate, land cover, and livestock density. These studies are valuable for identifying broad geographical risk patterns, but their purpose differs from district-month operational classification. Ecological niche suitability indicates where environmental conditions may favour disease occurrence, whereas the present system estimates risk for a specific district and month using concurrent and historical information.

2.3.4 Differences Between Existing Research and the Present Study

Existing studies demonstrate that livestock disease prediction benefits from environmental and spatial information. However, the literature is divided across several modelling traditions.

Statistical surveillance focuses on abnormal changes in case counts. Spatial epidemiology identifies clustering and geographical risk. Ecological niche models estimate environmental suitability. Machine-learning approaches attempt to classify disease occurrence from a larger set of predictors.

The present study draws elements from each of these areas but addresses a different operational problem: constructing a district-month risk estimate from sparse veterinary records while retaining temporal validity, uncertainty information, and model interpretability.

2.4 Existing systems products and techniques

2.4.1 Conventional Surveillance Techniques

Conventional surveillance systems generally collect disease notifications and summarise them according to time and location. Their primary advantage is that they are based on observed disease information and can provide an official epidemiological record.

Their limitation is that they are mainly reactive. An outbreak must normally be observed and reported before it becomes visible in the surveillance system. Statistical alerting methods can improve this process by identifying abnormal patterns, but they still depend heavily on consistent reporting histories.

2.4.2 Statistical Count Models

Poisson, negative-binomial, zero-inflated, and hurdle models are commonly used when epidemiological outcomes are represented as counts. These approaches can be useful when data contain many zero observations and when the main research question concerns outbreak counts.

For the present research, however, reliable monthly count severity was not consistently available across the complete district-month structure. The primary task was therefore treated as binary outbreak-risk classification rather than continuous count forecasting.

2.4.3 Ecological Niche and Spatial Risk Models

Maximum Entropy and related ecological niche techniques have been widely used for disease-suitability modelling. Their main strength is the ability to combine geographical occurrence data with environmental surfaces to identify areas that share characteristics with known outbreak locations.

Studies such as those by Li et al. [8] and An et al. [9] show that these methods can provide useful large-scale LSD risk maps.

Their limitation for the present objective is that spatial suitability alone does not directly represent month-specific district risk or recent epidemiological history.

2.4.4 Machine-Learning Classifiers

Machine-learning methods such as Logistic Regression, Random Forest, Gradient Boosting, artificial neural networks, and related ensemble techniques have increasingly been used for disease prediction.

Complex models can represent non-linear relationships and interactions, but this flexibility may increase overfitting risk when positive disease observations are limited. More complex models can also be more difficult to explain.

For the present study, Logistic Regression and regularised variants offer several advantages. They are suitable for binary outcomes, work effectively with structured tabular predictors, support class weighting and calibration, and retain a direct mathematical relationship between individual predictors and estimated log-odds.

A more complex benchmark can still be useful to determine whether additional predictive flexibility produces meaningful improvement.

2.4.5 Calibration and Decision Thresholds

A default classification threshold of 0.50 is not necessarily appropriate for rare-disease surveillance. When the cost of failing to identify an outbreak is greater than the cost of investigating a false-positive alert, a lower operating threshold may be justified.

Threshold selection should therefore be considered an operational decision rather than a fixed property of the machine-learning model. The present research evaluates this trade-off explicitly by examining how threshold changes affect recall and false-positive behaviour.

2.4.6 Explainability and Uncertainty Techniques

Many machine-learning systems produce a score without clearly communicating why that score was generated or how reliable it is.

Post-hoc explanation methods can provide useful information for complex models. However, when the final classifier itself is linear, exact model-based attribution provides a more direct explanation of how features affect log-odds.

Similarly, probability values alone should not be treated as complete representations of uncertainty. Conformal prediction provides an additional mechanism for indicating whether the available evidence supports one class strongly or whether more than one outcome remains plausible at the selected coverage level [16], [17].

Together, model-based interpretation and formal uncertainty analysis provide a stronger basis for evaluating disease-risk models than an unexplained binary alert alone. In the implemented system, explanation is delivered directly through linear log-odds contributions, while conformal uncertainty is retained as an evaluated research method rather than a live production output.

2.5 Gap synthesis and conceptual framework

The reviewed literature shows clear progress in livestock disease surveillance and predictive modelling. Traditional statistical systems provide established methods for detecting changes in disease counts [1] – [3]. FMD research demonstrates the importance of spatial, seasonal, livestock, and environmental context [4], [5]. LSD studies provide strong evidence that climatic, geographical, and livestock-density variables can contribute to disease-risk modelling [6] – [9]. Machine-learning research has also demonstrated the potential to classify disease occurrence from complex environmental data [7].

However, these areas are often treated separately.

First, much of the established outbreak-detection literature assumes relatively regular surveillance observations. This does not directly solve the problem of reconstructing sparse veterinary records into a consistent analytical time grid.

Second, ecological niche and spatial risk models are useful for identifying geographical suitability, but they do not necessarily provide operational district-month estimates that incorporate recent disease history.

Third, machine-learning studies may report strong predictive performance without always testing whether that performance remains stable across later years. Research on structured and temporally dependent data shows that validation design can strongly affect performance estimates [12], [15].

Fourth, rare-outbreak classification requires evaluation that reflects class imbalance. ROC-AUC alone may not adequately represent performance on the minority outbreak class, making PR-AUC and threshold-specific metrics important [13].

Finally, predictive performance alone is insufficient for veterinary decision support. A model intended to assist monitoring should communicate both the basis of its predictions and the uncertainty associated with them [16] – [18].

Based on this synthesis, the research gap can be expressed as follows:

There is limited evidence of an integrated district-level livestock disease-risk framework for Sri Lanka that reconstructs sparse surveillance information,

combines local climate, seasonality, livestock-related variables, spatial and temporal outbreak history and large-scale climate context, evaluates models using time-aware and year-separated leakage-aware procedures, and communicates both interpretable risk estimates and predictive uncertainty for FMD and LSD.

The conceptual framework developed to address this gap consists of five connected stages:

1. Surveillance reconstruction - Sparse disease records are converted into a consistent district-month analytical structure.

2. Spatio-temporal data integration - The disease records are combined with climate, seasonality, livestock-density, teleconnection, neighbouring outbreak, and historical outbreak variables.

3. Outbreak-risk modelling - Calibrated binary classifiers estimate the probability of outbreak occurrence for each district-month.

4. Reliability and interpretation - Time-aware or year-separated validation, class-sensitive metrics, threshold analysis, offline conformal uncertainty evaluation, and model-based feature attribution are used to assess and explain model behavior.

5. Decision-support delivery - The resulting risk information is delivered through the software component as advisory information for veterinary monitoring and prioritisation.

This framework positions the research between conventional surveillance and long-horizon disease forecasting. Its primary analytical function is to estimate district-month outbreak risk from available environmental and epidemiological information. Where future dates are requested by the software, historical same-calendar-month environmental conditions are used as climatological proxies; these outputs are therefore presented as risk projections rather than validated multi-month forecasts.

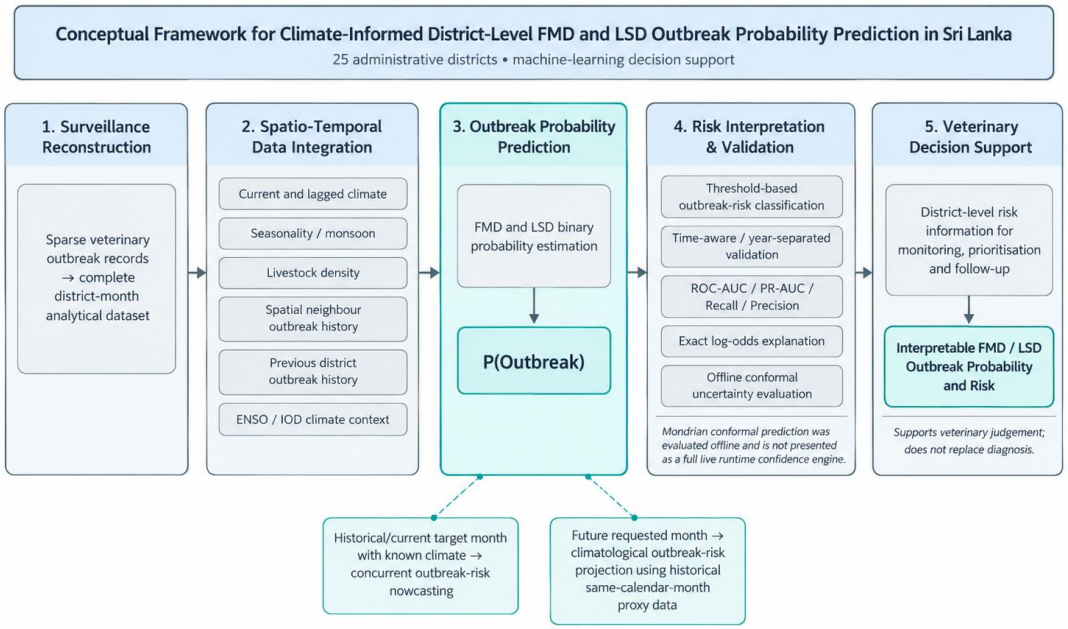


Figure 2 : Conceptual Framework of the Proposed Disease-Risk Approach

Table 3 : Comparison of Related Approaches

Study / Approach	**Disease**	**Main Data / Variables**	**Approach**	**Key Strength**	**Limitation Relative to This Study**
Farrington et al. [1]	Infectious disease surveillance	Historical case counts	Statistical outbreak detection	Established aberration-detection approach	Requires regular historical surveillance; limited environmental integration
Hutwagner et al. [2]	General surveillance	Routine surveillance counts	EARS	Rapid abnormal-pattern detection	Mainly count-based and reactive
González Gordon et al. [4]	FMD	Spatial and temporal outbreak studies	Scoping review	Demonstrates importance of space, seasonality and risk factors	Does not provide a single operational Sri Lankan model

Mielke and Garabed [5]	FMD	Temperature, humidity, environmental survival	Environmental persistence analysis	Supports environmental basis for climate features	Focuses on virus persistence rather than district-month classification
Allepuz et al. [6]	LSD	Climate, land cover, cattle density	Spatial regression	Identifies geographical risk areas	Primarily spatial rather than operational monthly risk
Afshari Safavi [7]	LSD	Meteorological and geospatial features	Multiple ML classifiers	Demonstrates ML potential for LSD occurrence	Different epidemiological setting and validation structure
Li et al. [8]	LSD	Climate, livestock density, land cover	Ecological niche modelling	Spatio-temporal Asian risk assessment	Environmental suitability rather than district-month nowcasting
An et al. [9]	LSD	Bioclimate, vectors, livestock density	Ecological niche modelling	Broad global risk mapping	Limited recent outbreak-history representation
Present study	FMD and LSD	Climate, seasonality, livestock, teleconnections, spatial and temporal outbreak history	Calibrated spatio-temporal classification / nowcasting	Integrates time-aware and year-separated validation, uncertainty and interpretation	Limited by sparse surveillance data and temporal instability, particularly for LSD

3 RESEARCH METHODOLOGY

3.1 Research design

This study followed an applied quantitative research design combined with software-system development. The research focused on constructing and evaluating a district-level livestock disease-risk nowcasting component for Foot-and-Mouth Disease (FMD) and Lumpy Skin Disease (LSD) in Sri Lanka.

The methodological process consisted of five main stages: preparation of district-month epidemiological datasets, integration of environmental and epidemiological predictors, model development, time-aware and year-separated evaluation, and integration of the selected models into the software system.

The primary analytical task was binary outbreak-risk classification at district-month level. When actual climatic information for the assessed month was available, the model estimated the probability of outbreak occurrence for that same district-month and therefore operated as a concurrent nowcasting classifier. The software also supports future-date risk projection; however, these projections use historical same-calendar-month conditions when actual future climate data are unavailable and are treated as climatological scenario estimates rather than validated multi-month forecasts.

The research used a two-stage modelling architecture. Stage 1 estimates outbreak occurrence probability, while Stage 2 is conditionally evaluated when the Stage 1 probability reaches the operational threshold. FMD Stage 2 provides advisory severity classification, whereas the LSD Stage 2 model is more appropriately interpreted as a quiet-period false-alarm suppressor than as a validated active-wave severity discriminator. Interpretability is provided in the runtime through exact log-odds decomposition, while Mondrian conformal prediction was investigated separately during offline uncertainty analysis.

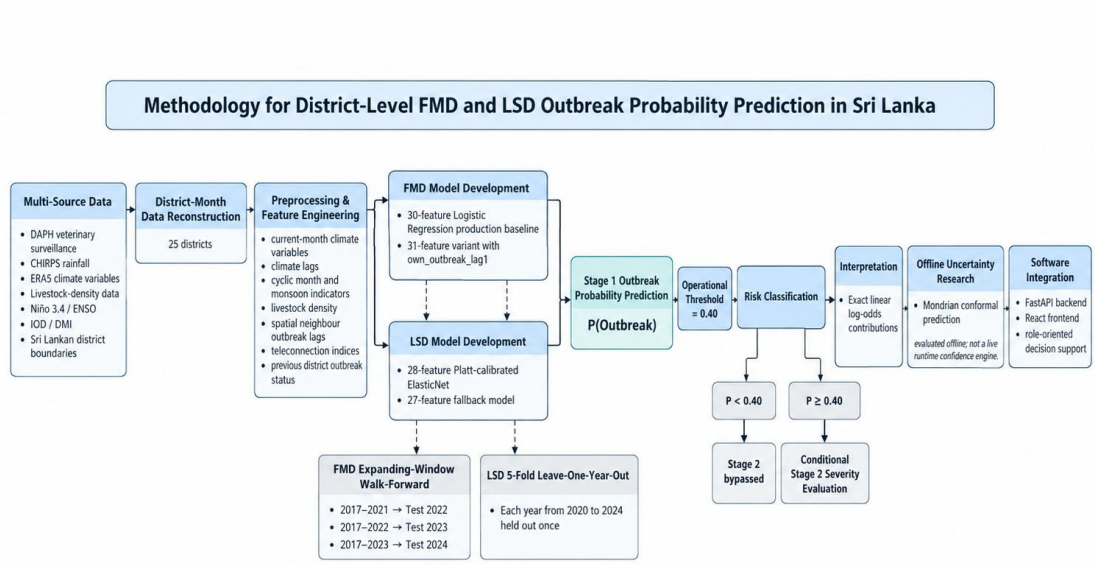


Figure 3 : Research Methodology and Model-Evaluation Pipeline

3.2 Data participants or study materials

This study did not involve direct human participants. The main study materials were historical veterinary surveillance records and environmental datasets aligned at district-month level.

The surveillance information was derived from DAPH epidemiological materials [19], while rainfall information was obtained from CHIRPS [20] and atmospheric variables were derived from ERA5 [21].

The FMD processed dataset contained **2,400 district-month observations** covering 25 districts from 2017 to 2024. The LSD dataset contained **1,500 district-month observations** covering 2020 to 2024. Outbreak-positive observations represented approximately 12.75% of the FMD dataset and 7.67% of the LSD dataset, indicating a strong class-imbalance problem.

The epidemiological records were combined with environmental and contextual variables. These included:

- rainfall and rainfall-derived features;
- humidity and wind-related variables;

- seasonal and monsoon indicators;
- livestock-density variables;
- Niño 3.4 and Indian Ocean Dipole indices;
- previous outbreak activity in the same district;
- outbreak activity in neighbouring districts.

The FMD production pipeline contains a 30-feature baseline model and an enhanced 31-feature variant. The additional feature in the latter is own_outbreak_lag1, representing outbreak occurrence in the same district during the previous month. The LSD production pipeline uses a calibrated 28-feature ElasticNet model when verified previous-month outbreak information is available and falls back to a 27-feature model otherwise.

Missing and unavailable feature information was handled through the preprocessing and runtime data-provider logic. Model scaling parameters were fitted from training data before evaluation. During software inference, an exact district-month record is used where available. For future or unavailable periods, the provider first attempts to use the most recent historical record for the same district and calendar month; if that is unavailable, district-level or national historical medians are used as fallback values. Because these fallback procedures draw on historical summary information rather than a fully date-bounded expanding window, they are treated as a limitation when interpreting future projections.

3.3 Experimental or study procedure

The experimental procedure began by reconstructing disease-surveillance data into complete district-month grids. Epidemiological records were aligned with environmental information according to district and month.

Feature engineering was then applied to represent both current conditions and recent disease history. Seasonal patterns were represented using cyclic month variables and monsoon indicators. Lagged rainfall and climate variables were used to capture recent environmental conditions, while spatial lag variables represented outbreak

activity in neighbouring districts. Recent outbreak activity within the target district was included through temporal autocorrelation features.

For FMD, the modelling process progressed from earlier baseline configurations to a 30-feature climate-spatial production baseline and a 31-feature variant incorporating own_outbreak_lag1. The 31-feature variant produced stronger discrimination and was therefore retained as an important, optimized research configuration, while the 30-feature model remained the default production baseline because it does not depend on verified previous-month target information.

For LSD, the primary model was an ElasticNet Logistic Regression model wrapped within CalibratedClassifierCV. A LightGBM configuration was retained as a benchmark, but the calibrated ElasticNet model was used as the main production model.

The validation strategy differed between the two diseases according to the available data. FMD was evaluated using expanding-window walk-forward folds in which earlier years were used to predict later held-out years, including a final 2024 test fold. LSD was evaluated using Leave-One-Year-Out cross-validation across 2020–2024, with each complete calendar year held out in turn. The LSD procedure therefore provides year-separated out-of-sample evaluation rather than a strictly forward-only chronological forecast design.

For FMD, threshold analysis was also performed because a default probability threshold of 0.50 was not necessarily suitable for an imbalanced veterinary-surveillance problem. A threshold of 0.40 was examined to improve outbreak recall.

After probability estimation, the research evaluated Mondrian conformal prediction as an offline uncertainty-quantification method, particularly for the FMD model. Runtime model explanations were generated using exact linear log-odds attribution. These explanation outputs, together with thresholded risk and conditional severity information, were subsequently integrated into the backend service and frontend decision-support interface.

3.4 Measures and evaluation metrics

Model performance was evaluated using metrics appropriate for imbalanced binary classification.

ROC-AUC was used to measure the ability of the model to rank outbreak observations above non-outbreak observations across different thresholds.

PR-AUC was included because the positive outbreak class is relatively rare. Precision-recall analysis provides a more direct view of model behaviour on the minority class.

Recall measured the proportion of observed outbreaks correctly identified by the model:

$$Recall = \frac{TP}{TP + FN}$$

where ***TP*** represents true positives and ***FN*** represents false negatives.

Precision measured the proportion of predicted outbreak alerts that corresponded to actual outbreak observations:

$$\text{Precision} = \frac{TP}{TP + FP}$$

The classification threshold was analysed because lowering the threshold can increase outbreak detection while also increasing false-positive alerts.

For LSD, performance was examined across complete held-out calendar-year folds under the LOYO protocol. Mean ROC-AUC and PR-AUC values were reported together with fold-level results because a single aggregate value can conceal substantial variation between individual years.

Mondrian conformal prediction coverage was evaluated separately as an offline uncertainty measure. In the FMD audit, class-conditional conformal prediction achieved approximately 94.9% empirical coverage for outbreak cases at the selected nominal coverage setting. This result is treated as research validation of the method

and not as evidence that the current production endpoints return full conformal confidence sets.

3.5 Data analysis

The analysis focused on three main questions: predictive discrimination, temporal generalisation, and operational usefulness.

First, FMD model configurations were compared to determine whether adding recent district-level outbreak history improved classification performance. The 31-feature variant achieved stronger discrimination than the 30-feature baseline, allowing the contribution of recent district-level target autocorrelation to be assessed separately from the more generally deployable baseline feature set.

Second, LSD performance was examined across each held-out year under the five-fold LOYO protocol. The deployed Platt-calibrated ElasticNet model achieved a mean ROC-AUC of 0.6136 and mean PR-AUC of 0.2699. Fold-level results were retained in the analysis because the relatively small number of positive observations and differences between years make the mean score alone insufficient for judging model reliability.

Third, threshold analysis was used to examine the trade-off between recall and false-positive alerts. In a veterinary-surveillance setting, failing to flag an outbreak can be more consequential than generating an additional advisory alert. Therefore, threshold selection was evaluated in relation to outbreak detection rather than overall accuracy alone.

Interpretability analysis for the deployed linear models used exact log-odds decomposition. For a standardized feature $\boldsymbol{k}$, its contribution to the model decision score is:

$$Contribution_k = \beta_k x_{k,\text{scaled}}$$

where β_k is the fitted model coefficient and x_k,scaled is the **standardized** feature value supplied to the classifier. The model decision score can therefore be reconstructed as:

$$z = \beta_0 + \sum_{k=1}^{p} \beta_k \, x_{k,\text{scaled}}$$

and converted to outbreak probability using the logistic sigmoid. This provides a direct explanation of the fitted linear model rather than relying on an approximate post-hoc method during runtime.

3.6 Ethics privacy security and responsible AI

The study did not involve experimentation on animals or direct collection of personal information from human participants. The research used historical disease-surveillance and environmental data.

The disease-risk outputs are intended only as **advisory decision support**. The model does not diagnose individual animals and does not automatically determine vaccination, quarantine, movement restrictions, or other veterinary interventions. Final decisions remain the responsibility of veterinary professionals and relevant authorities.

Because disease-risk information could potentially be misinterpreted or used inappropriately, the application includes role-based access controls for relevant system functions. This helps reduce the risk of unauthorised access or misuse of disease-risk information.

Responsible AI considerations included transparent reporting of model limitations, clear separation between statistical risk estimation and clinical diagnosis, and interpretable prediction outputs. Weak or inconsistent results were retained in the analysis rather than being concealed by aggregate metrics. Formal uncertainty methods were evaluated during the research, while operational outputs were deliberately presented as advisory risk information rather than guaranteed predictions.

A further ethical consideration concerns data provenance. Formal independent certification for some DAPH-derived surveillance records was not available during the study. This limitation is reported explicitly rather than presenting the data source as more certain than the available evidence supports.

3.7 Reproducibility and research artefacts

Reproducibility was supported by preserving trained model artefacts, source code, evaluation scripts, and automated test evidence.

The final model registry included:

- FMD Stage 1 baseline Logistic Regression;
- FMD 31-feature Logistic Regression;
- FMD Stage 2 Random Forest severity model;
- LSD 28-feature calibrated ElasticNet model;
- LSD 27-feature fallback model.
- LSD Stage 2 Logistic Regression severity/suppression model.

The software implementation used Python 3.12 for the backend and TypeScript for the frontend. Major libraries included Scikit-Learn, LightGBM, Imbalanced-Learn, Pandas, NumPy, Joblib, FastAPI, and Pydantic.

Automated testing covered both model-related backend behaviour and the frontend interface. The final repository test record contained **862 passing automated tests**, consisting of 371 backend tests and 491 frontend tests, with no recorded failures in the final test run.

The component exposes disease-specific inference routes under /api/v1/risk-forecasting. The primary prediction endpoints are /predict/fmd and /predict/lsd, while /forecast/fmd and /forecast/lsd generate multi-district climatological risk projections for a requested calendar month. The backend also maintains supporting routes for forecast records, advisories, notifications, and follow-up workflows.

These artefacts provide a traceable link between the research methodology, trained models, reported results, and deployed software component.

4 PRODUCT AND SOLUTION DESIGN

4.1 Requirements

The disease-risk component was designed as a decision-support module within the wider livestock-health platform. Its main purpose is to transform district-level disease and environmental information into understandable outbreak-risk outputs for veterinary use.

The functional requirements were derived from the research objectives and from the need to integrate the modelling workflow into the complete application. The main requirements were to:

- allow the user to select a disease, district, month, and year;
- generate a district-month outbreak-risk probability;
- apply the relevant decision threshold and produce a risk classification;
- provide conditional severity information where the Stage 2 model is applicable;
- provide interpretable feature contributions for individual predictions;
- communicate data quality, model limitations, and role-appropriate risk information to users;
- return the canonical list of Sri Lankan districts;
- provide historical district information to authorised users;
- integrate prediction results into the wider frontend application.

The principal disease-specific prediction routes are /api/v1/risk-forecasting/predict/fmd and /api/v1/risk-forecasting/predict/lsd. Separate /forecast/fmd and /forecast/lsd routes generate district-wide risk projections for a requested month. Supporting endpoints are used for forecast records, advisories, follow-ups, and related operational workflows.

The main non-functional requirements concerned reliability, security, maintainability, usability, and reproducibility. The component needed to produce

consistent outputs from saved model artefacts, validate incoming requests, restrict sensitive historical information according to user role, and present model results in a form understandable to non-technical users.

Because the system is intended for advisory use, another important requirement was to avoid presenting model outputs as confirmed veterinary diagnoses.

4.2 Component architecture within the complete solution

The disease-risk component follows a layered architecture connecting the user interface, backend API, forecasting services, and trained model artefacts.

At the presentation layer, the React/TypeScript frontend allows users to select the target district, date, and disease and then visualises the returned risk information. The interface includes district-level risk views, probability indicators, trends, and explanation information.

Requests from the frontend are sent to the FastAPI backend. The backend is responsible for request validation, authentication and role-based access control, and routing prediction requests to the relevant forecasting service.

The forecasting service constructs the required feature vector from stored district-month information, static attributes, current or proxy climatic values, and historical lag features. For FMD, the service can route between the 30-feature baseline and 31-feature autocorrelation variant depending on the availability of verified previous-month outbreak status. The LSD service similarly uses either the 28-feature primary model or the 27-feature fallback model. After Stage 1 probability estimation, the configured threshold is applied, and Stage 2 is conditionally evaluated. The service also generates exact log-odds feature contributions and records data-quality information describing whether exact observations or historical proxy values were used.

The resulting response is returned to the frontend as structured JSON and displayed as advisory information.

The simplified component flow is:

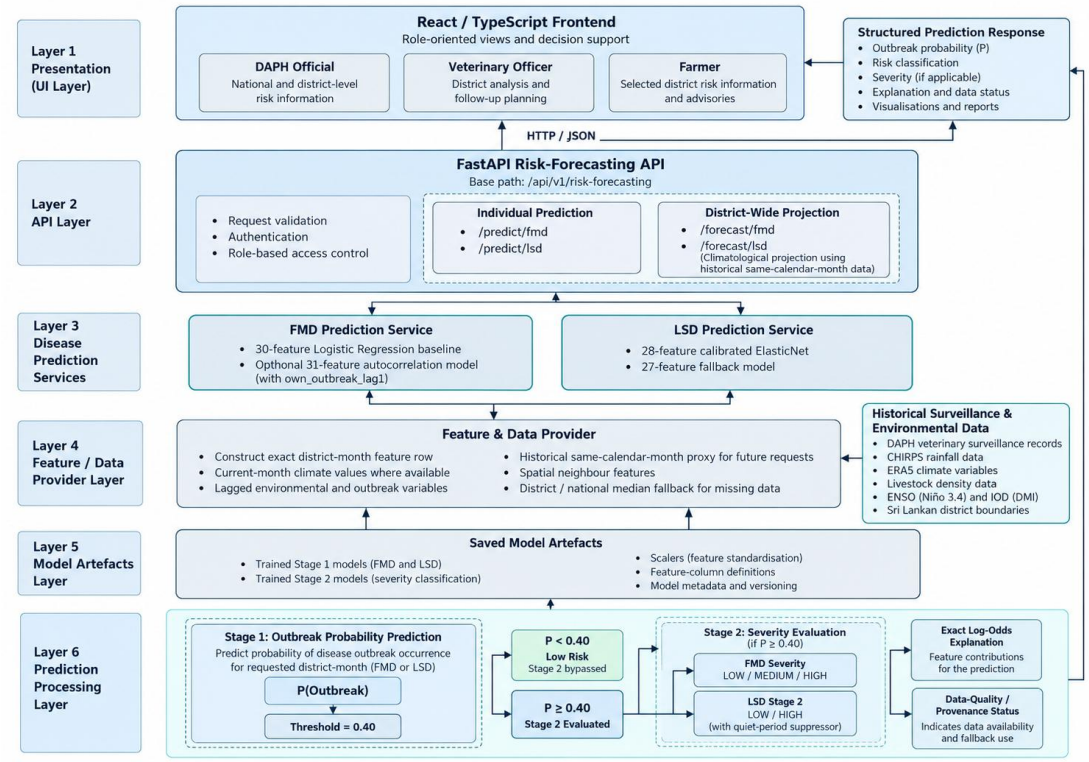


Figure 4 : Architecture of the Disease Forecasting Component

Within the complete R26-SE-013 solution, the disease-risk module operates as a separate research component while sharing the wider platform infrastructure. This separation reduces direct dependency between components and allows the forecasting service to be tested and maintained independently.

The implementation uses Python 3.12 and FastAPI on the backend and React with TypeScript on the frontend. Model-related libraries include Scikit-Learn, LightGBM, Pandas, NumPy, Imbalanced-Learn, and Joblib.

4.3 Design alternatives and decisions

Several design decisions changed during the development of the component as the research findings became clearer.

Model complexity versus interpretability

Tree-based ensemble models were considered because they can represent complex non-linear relationships. However, the final FMD Stage 1 model uses Logistic

Regression, while the primary LSD model uses regularised ElasticNet Logistic Regression with calibration.

This choice provided a balance between predictive performance, interpretability, and reproducibility. Linear models also allow feature influence to be calculated directly from the fitted coefficients, which was useful for producing understandable risk explanations.

Forecasting versus nowcasting

The original project concept used the broader term disease forecasting. During implementation, the available feature pipeline required several climatic variables from the month being evaluated. When those target-month observations are available, the model therefore operates as a concurrent district-month nowcasting classifier.

The software also allows a user to request a future year and month. In this situation, the backend does not obtain future weather forecasts. Instead, it retrieves the latest historical observation for the same district and calendar month and uses that information as a climatological proxy. The future-facing system function is therefore described in this thesis as climatological risk projection rather than validated multi-month forecasting.

SHAP versus direct linear attribution

SHAP was used during offline exploratory research to examine global feature behavior and produce model-interpretation plots. However, SHAP is not executed during live API inference. Because the deployed Stage 1 classifiers are linear models, the production services use exact coefficient-based log-odds decomposition to explain individual predictions. This provides deterministic explanations with lower runtime overhead while remaining directly connected to the fitted decision function.

Bootstrap uncertainty versus conformal prediction

Bootstrap-based uncertainty intervals were examined during earlier research stages but produced inadequate empirical coverage. Mondrian class-conditional conformal

prediction was subsequently evaluated offline and showed substantially improved coverage for FMD outbreak cases. However, the current production endpoints do not expose the full conformal calibration procedure at runtime. Conformal prediction is therefore retained as an evaluated research method and a basis for future production uncertainty support rather than being described as a fully deployed live feature.

Monolithic processing versus service-based architecture

A notebook- or script-based design would have been sufficient for experimentation but would not support system integration effectively. The final design therefore exposes forecasting functionality through a FastAPI service, allowing the model pipeline to operate independently from the user interface.

These changes reflect the movement from an experimental research prototype toward a deployable software component.

4.4 Detailed design

The prediction workflow begins when a user selects the disease, district, month, and year through the frontend interface. The frontend then sends the request to the appropriate disease-specific endpoint under /api/v1/risk-forecasting.

For individual FMD prediction, the request is directed to:

POST /api/v1/risk-forecasting/predict/fmd

while LSD prediction is handled through:

POST /api/v1/risk-forecasting/predict/lsd

The backend validates the incoming request and passes the target district and date to the corresponding forecasting service. The forecasting service then requests the required feature row from the data-provider layer.

If an exact district-month record exists, the associated feature values are used directly. These can include target-month rainfall, humidity, wind speed, teleconnection indices, calendar and monsoon variables, livestock-density information, and lagged climate and outbreak-history features. If the requested period is in the future and no exact record exists, the provider retrieves the latest

available historical observation for the same district and calendar month. Where this is not possible, district or national historical medians are used as fallback values.

The FMD prediction service normally uses the 30-feature Logistic Regression baseline. When verified previous-month outbreak information is available and the enhanced model is requested, the 31-feature variant adds own_outbreak_lag1. The LSD service uses the calibrated 28-feature ElasticNet model when verified previous-month outbreak status is available and automatically switches to its 27-feature fallback model otherwise.

The selected feature vector is transformed using the scaler associated with the trained model before Stage 1 probability estimation. The resulting probability represents the estimated risk of an outbreak for the requested district-month.

An operational threshold of 0.40 is then applied. When the Stage 1 probability is below this threshold, Stage 2 is bypassed. When the threshold is met or exceeded, the appropriate Stage 2 model is evaluated. FMD uses a Random Forest severity classifier that can return LOW, MEDIUM, or HIGH severity categories. The LSD Stage 2 model returns a binary severity-oriented output but is treated operationally as a quiet-period false-alarm suppressor because it has not demonstrated reliable active-wave severity discrimination.

For model interpretation, the backend calculates feature-level log-odds contributions directly from the fitted linear classifier. Positive contributions indicate features pushing the decision score towards higher outbreak risk, while negative contributions reduce the score. The reconstructed decision score and probability can therefore be checked against the classifier output.

The backend also returns information about the provenance of the feature row used during prediction. This allows the frontend to distinguish an exact historical/current-period result from a result generated using a historical same-month proxy or median fallback.

In addition to individual prediction endpoints, /forecast/fmd and /forecast/lsd iterate across all 25 districts for a requested calendar month. For dates beyond the available

surveillance period, these routes produce climatological risk projections using historical same-month proxy conditions rather than future meteorological forecasts.

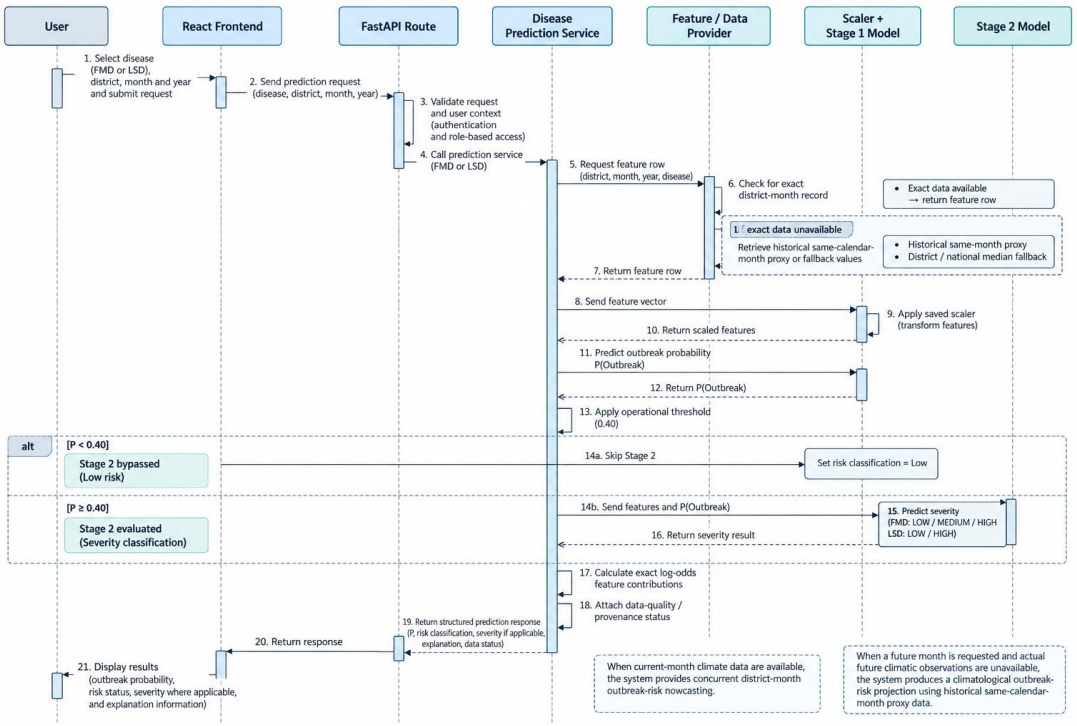


Figure 5 : End-to-End Sequence of an FMD or LSD Outbreak Probability Prediction Request

The structured backend response is then rendered in the React interface using role-appropriate risk indicators, severity information, explanation cards, historical context, and operational guidance.

4.5 Standards and professional practice

The disease-risk component was designed using established software-engineering practices together with principles appropriate for responsible machine-learning decision support.

Separation of concerns

The frontend, API routing layer, forecasting services, data-provider logic, persistence layer, and trained model artefacts are maintained as separate parts of the application. This separation reduces coupling between the user interface and the prediction logic and allows individual parts of the component to be maintained and tested independently.

API-based integration

The forecasting functionality is exposed through structured REST-style endpoints implemented with FastAPI. Pydantic models are used to validate request and response data, helping maintain predictable contracts between the frontend and backend. Disease-specific prediction and forecast routes also allow the FMD and LSD workflows to be handled independently while remaining within the same application architecture.

Security and access control

Authentication and role-based access control are used to restrict access to operational and surveillance-related functions. Protected operations follow fail-closed behaviour when valid user context is unavailable. The wider backend also uses token-based authentication and secure password-hashing mechanisms. Forecast-related records include safeguards such as idempotency and immutable prediction information to reduce accidental duplication or unauthorised modification of recorded decisions.

Testing and quality assurance

The component was supported by automated backend and frontend testing. The final repository test evidence recorded 862 passing automated tests across the backend and frontend, with no failures in the final recorded test run. Testing covered the forecasting component as well as shared backend and frontend behaviour.

Responsible decision support

The system was designed to support veterinary judgement rather than replace it. Predictions represent district-level statistical risk and should not be interpreted as confirmed diagnosis of an individual animal. Stage 2 severity information is also presented as advisory output, with additional limitations applied to the LSD severity model.

The interface and backend communicate limitations where relevant. Future-date risk projections indicate when historical same-calendar-month proxy data have been used instead of observed target-period data. Runtime explanations are generated through exact log-odds feature contributions, while formal Mondrian conformal uncertainty

analysis is treated as an offline research result rather than presented as a fully deployed live confidence mechanism.

Traceability and reproducibility

The project preserves trained model artefacts, scalers, feature-column definitions, source code, evaluation scripts, testing evidence, and documented model configurations. The system also records data-quality information associated with predictions, allowing a result based on an exact district-month record to be distinguished from one produced using historical proxy or fallback data. These practices provide traceability between the research experiments and the behaviour of the implemented software.

5 IMPLEMENTATION INTEGRATION AND TESTING

5.1 Development environment and technology stack

The disease-risk component was implemented as part of the wider R26-SE-013 web application. The backend was developed in **Python 3.12** using **FastAPI**, while the frontend was implemented using **React 18** with **Vite** and **Tailwind CSS**. The backend uses **Pydantic** for request and response validation and **Uvicorn** as the ASGI server during application execution.

The machine-learning implementation was built mainly with **Scikit-Learn**, together with Pandas, NumPy and Joblib for data processing and model persistence. Imbalanced-Learn was used during model-development experiments involving class imbalance, particularly SMOTE for the FMD Stage 2 severity model. Tree-based libraries including LightGBM, XGBoost and CatBoost were used during benchmarking rather than as the final Stage 1 production models.

The frontend and backend were kept separate so that the user interface did not depend directly on model files or data-processing logic. Trained estimators, scalers and feature-column definitions were saved as model artefacts and loaded by the backend services at runtime. This avoided re-training models whenever a prediction request was received.

The component was maintained within the central R26-SE-013 repository. Its main backend implementation is located under backend/components/risk_forecasting/, while the related frontend implementation is maintained under frontend/src/features/RiskForecasting/. The latest audited project structure also separates service logic, repositories and external integration boundaries within the forecasting component.

5.2 Component implementation

The implementation was divided into backend prediction services and a role-oriented frontend interface.

5.2.1 Backend implementation

The backend forecasting component is organised around routes, schemas, services, repositories and integration modules. The main service layer contains separate logic for FMD and LSD prediction, allowing the two diseases to use different models and fallback behaviour without combining their implementation into a single prediction routine.

For an FMD prediction, the backend loads the required feature values, applies the associated scaler and passes the transformed vector to the selected Logistic Regression model. The default production path uses the 30-feature model. When verified previous-month disease information is available and the enhanced model is requested, the 31-feature model can additionally use own_outbreak_lag1. For LSD, the service uses the calibrated 28-feature ElasticNet model when the previous-month outbreak value is available and switches to the 27-feature fallback model when that value cannot be verified.

Individual predictions are exposed through the following endpoints:

- POST /api/v1/risk-forecasting/predict/fmd
- POST /api/v1/risk-forecasting/predict/lsd

The component also exposes:

- POST /api/v1/risk-forecasting/forecast/fmd
- POST /api/v1/risk-forecasting/forecast/lsd

The forecast routes generate district-level risk results across all 25 districts for a requested calendar month. For future periods, the backend does not obtain future meteorological forecasts. Instead, it uses the latest historical values available for the same district and calendar month as a climatological proxy. The response records whether the input came from an exact record, historical proxy or fallback median so that the origin of the prediction can be distinguished.

After Stage 1 produces an outbreak probability, the common operational threshold of **0.40** is applied. If the probability is below this threshold, Stage 2 is bypassed. For FMD predictions that meet or exceed the threshold, the saved Random Forest

severity model can return LOW, MEDIUM or HIGH advisory severity. LSD also contains a Stage 2 Logistic Regression model, but its output is treated more cautiously because the model mainly functions as a quiet-period suppressor rather than as a reliable discriminator of severity during active outbreak periods.

Runtime model interpretation is produced using exact linear log-odds contributions. The prediction services do not execute SHAP during live inference. Instead, each feature contribution is calculated from the fitted coefficient and the standardised feature value. SHAP was retained for offline exploratory analysis and visualisation during model research.

The backend also contains supporting services for forecast records, advisories, notifications and veterinary follow-up workflows. Forecast decisions can be persisted as immutable records, while idempotency checks are used to prevent unintended duplication of equivalent requests.

5.2.2 Frontend implementation

The user interface was developed as a React-based feature area within the wider livestock-health platform. Rather than presenting the same level of information to every user, the implementation contains role-oriented interfaces for DAPH officials, veterinary officers and farmers.

The DAPH-facing views provide national and district-level risk information together with follow-up and data-quality functions. Veterinary views focus on district surveillance, assigned investigations and advisory review. Farmer-oriented views present simplified risk and practical guidance without exposing the same technical model detail available to professional users.

The interface presents model outputs such as outbreak probability, risk status, conditional severity and explanatory feature contributions. Technical interpretation is reduced for non-technical roles, while veterinary and DAPH users can access more detailed information.

An important implementation decision was to retain the distinction between a statistical risk estimate and a clinical diagnosis. The frontend therefore operates at district level and does not convert model probability into a diagnosis for an

individual animal. Additional interface guardrails are used where the underlying model has known limitations, including the LSD Stage 2 severity result.

5.3 Integration with the other components

The disease-risk component was integrated into the common R26-SE-013 platform rather than deployed as an isolated research application. Integration takes place mainly through the shared FastAPI backend, authentication and role information, frontend application structure, database services and defined API contracts.

The forecasting module remains logically separated from the other research components. This was important because its inputs and outputs operate at district-month level, whereas other components address different tasks such as animal-level assessment, diagnosis or geospatial monitoring. Keeping the prediction logic within its own service boundary reduced unnecessary dependencies between independently developed research components.

Where cross-component information is required, integration is performed through explicit interfaces rather than by allowing one component to access another component's model implementation directly. One example is the outbreak-status endpoint, which exposes disease and district information for use by other parts of the overall system. The verified backend includes:

GET /outbreak-status/{disease}/{district}/{year}/{month}

as an integration bridge for related system components.

The component also participates in the wider operational workflow. Forecast results can be recorded and used to support advisory generation, notification processes and veterinary follow-up activities. These workflows connect risk estimation with the wider platform while retaining the original model output as a traceable forecast decision.

Role-based access is preserved during integration. DAPH officials can access broader surveillance and follow-up functions, while veterinary users operate within

relevant professional workflows. Restricted operations are not exposed solely because a user can access another part of the platform.

This approach allowed the individual research component to remain independently testable while still contributing its outputs to the complete livestock-health system.

5.4 Verification and software testing

Verification was carried out at both backend and frontend levels. Automated testing was used to check prediction behaviour, route handling, authentication, operational workflows and frontend rendering.

The risk-forecasting backend contains dedicated tests for model loading, prediction behaviour, decision thresholds and fallback logic. Additional backend tests cover authentication, operational services, route permissions, database behaviour and security-related functions. The frontend was tested using **Vitest** and **React Testing Library**, including role-based views, API interactions, navigation and component behaviour.

Table 4 : Test execution details

Test Area	**Tests Passed**
Backend	371
Frontend	491
Total	**862**

All **862 automated tests passed** in the final recorded execution, with no failures or errors. Of the backend tests, 27 specifically targeted the risk-forecasting component, while the remaining backend tests covered shared authentication, operational workflows, routes, database behavior and security functions used by the wider application.

Testing also considered access-control behavior and cross-role workflows. The integration guidance requires checks for authorized DAPH access, veterinary access, denial of unauthorized roles, invalid user context and cross-component behavior.

This was important because successful model execution alone would not demonstrate that predictions were being exposed safely through the complete application.

Model-performance validation is reported separately in Chapter 6. The purpose of the testing discussed here is to verify that the implemented software behaves according to its intended contracts rather than to repeat the statistical evaluation of the models.

5.5 Deployment and operation

The completed component operates through the shared web-application architecture. FastAPI hosts the backend prediction and operational services, while the React frontend communicates with the backend through HTTP requests and renders the returned district-level risk information.

At application startup, the required trained model artefacts, scalers and feature definitions are made available to the prediction services. A user request is validated by the backend before the relevant disease service is invoked. Prediction output is returned as structured data for presentation in the interface, while selected operational results can also be persisted as forecast decision records.

The implementation contains fallback behaviour intended to prevent unsupported data from being presented as if it were exact. When a requested future period has no matching district-month observation, the data provider identifies that historical proxy information has been used. If the required previous-month outbreak state is unavailable, the services can also switch from the target-autocorrelation models to their corresponding baseline or fallback models.

The current implementation should therefore be understood as an operational research system rather than a fully validated national production deployment. Wider deployment would require continued surveillance-data updates, external operational validation, monitoring of model performance over time, secure management of production credentials, and appropriate institutional governance.

This distinction is important because successful software integration and automated testing demonstrate technical readiness of the research component, but they do not by themselves establish clinical or national-level operational validation.

6 RESULTS AND VALIDATION

6.1 Research results

The research produced separate Stage 1 outbreak-risk models for FMD and LSD. Because the two diseases differed in data volume, prevalence and temporal behaviour, their results were evaluated separately rather than combined into a single overall performance measure.

6.1.1 FMD outbreak-risk results

For FMD, the 30-feature Logistic Regression model served as the production baseline, while a 31-feature variant additionally included own_outbreak_lag1 to represent outbreak occurrence in the same district during the previous month.

Table 5 : FMD Stage 1 model results

Model	Evaluation	ROC-AUC	PR-AUC	Recall at 0.40	Precision	F1
30-feature baseline	Walk-forward mean, 2022–2024	0.7833	0.3773	–	–	–
30-feature baseline	Held out 2024	0.7570	0.5461	65.4%	50.5%	0.570
31-feature autocorrelation variant	Walk-forward mean, 2022–2024	0.8120	0.4698	–	–	–
31-feature autocorrelation variant	Held out 2024	0.8166	0.6968	67.9%	53.0%	0.596

The enhanced 31-feature model produced better discrimination than the 30-feature baseline. In particular, the mean walk-forward ROC-AUC increased from 0.7833 to 0.8120 after adding recent district-level outbreak history. On the final 2024-fold, the enhanced model correctly identified 53 of the 78 observed outbreak district-months at the selected threshold of 0.40.

Threshold selection also affected the practical behaviour of the FMD classifier. A default threshold of 0.50 produced lower outbreak detection, while reducing the operational threshold to 0.40 increased recall. This supported the use of a lower threshold in the decision-support setting, where missing an outbreak period was considered more important than avoiding every false-positive alert.

The 31-feature result should not, however, be interpreted as evidence that previous outbreak occurrence causes a future outbreak. The improvement shows predictive value from recent disease history, but the relationship remains an association within the available surveillance data.

6.1.2 LSD outbreak-risk results

For LSD, the final production model was a 28-feature ElasticNet Logistic Regression classifier with Platt probability calibration. Performance was assessed using five Leave-One-Year-Out folds covering 2020–2024.

Table 6 : LSD model-development results

Model configuration	**Mean LOYO ROC-AUC**	**Mean LOYO PR-AUC**
23-feature climate-oriented baseline	0.5308	0.1276
27-feature spatial model	0.5559	0.1657
28-feature model with `own_outbreak_lag1`	0.6034	0.1876
28-feature ElasticNet	0.6167	0.2689
28-feature Platt-calibrated ElasticNet	0.6136	0.2699

Adding spatial outbreak information and recent district-level outbreak history improved performance, while ElasticNet regularization produced a further improvement in PR-AUC. The small reduction in ROC-AUC after Platt calibration, from 0.6167 to 0.6136, was accepted because calibration was introduced to improve the interpretation of probability outputs rather than to maximize ranking performance alone.

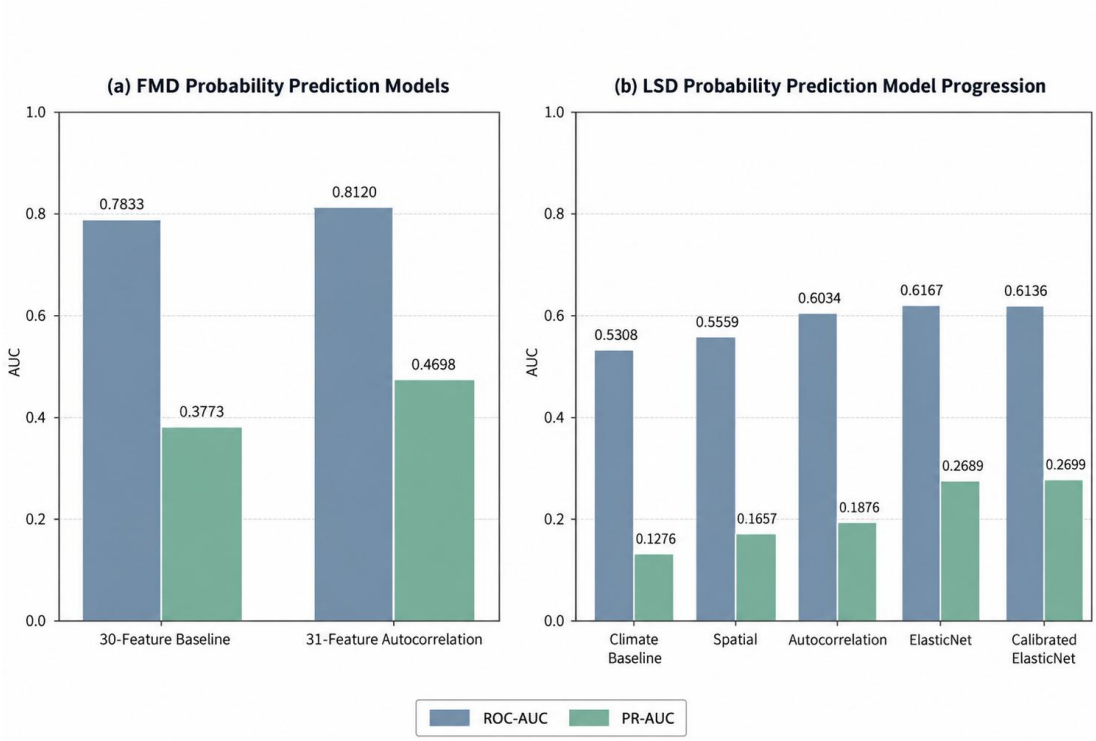


Figure 6 : Comparison of FMD and LSD Outbreak Probability Prediction Performance

The LSD results were weaker than the FMD results. This is important because the LSD dataset contained only five years of observations and relatively few positive district-months. The result therefore provides evidence of some predictive signal but does not support a claim of consistently strong performance across all years.

6.1.3 Uncertainty and interpretability results

Mondrian class-conditional conformal prediction was evaluated offline for the FMD classifier. At the selected nominal coverage level, the analysis produced approximately **95.3% overall empirical coverage** and **94.9% coverage for**

outbreak observations. Around 41% of the evaluated cases produced an informative single-class prediction set, while the remaining cases produced a two-class uncertainty set.

These results showed that class-conditional conformal prediction could provide useful uncertainty information for the FMD model. However, the conformal procedure is treated as an evaluated research result rather than as a fully deployed runtime feature.

For interpretability, the final runtime implementation uses exact linear log-odds decomposition. Each prediction can therefore be related directly to the fitted coefficients and transformed input values. SHAP was used during offline exploratory analysis for global feature interpretation, while the deployed API uses the deterministic log-odds calculation for individual predictions.

6.2 Product and technical results

The research models were successfully incorporated into the web-based decision-support component rather than remaining only as experimental notebooks or standalone model files.

The implemented backend supports separate FMD and LSD prediction routes and district-wide forecast routes under /api/v1/risk-forecasting. During inference, the service selects the appropriate model according to the disease and availability of previous-month surveillance information. FMD can use either the 30-feature baseline or the 31-feature autocorrelation variant, while LSD can switch between its 28-feature primary model and 27-feature fallback model.

The implementation also handles unavailable future-period information explicitly. If an exact district-month row is unavailable, the system can use the latest historical observation for the same calendar month and reports the result as a historical proxy rather than silently presenting it as observed future climate information.

The frontend provides role-oriented views for veterinary and DAPH users and presents district-level probability, risk status, conditional severity, trends and

explanatory information. Model outputs remain separated from individual-animal diagnosis.

Automated testing provided additional technical validation. The consolidated research evidence recorded **862 passing automated tests**, consisting of 371 backend tests and 491 frontend tests, with no failures in that recorded execution. The testing covered forecasting behaviour together with shared authentication, route handling and frontend functionality.

These results show that the research models were not only evaluated statistically but were also implemented as reusable software services within the wider platform.

6.3 User or stakeholder validation

The intended users of the component include veterinary officers and DAPH officials, with different levels of access to district-level risk information and operational functions.

The implemented interface reflects these roles. Veterinary users are provided with district-oriented surveillance information and follow-up functions, while DAPH-oriented views provide broader district and national-level risk summaries, data-quality information and oversight functions. The component therefore demonstrates that the research outputs can be presented in forms that correspond to different professional responsibilities.

However, no completed formal stakeholder study, structured usability questionnaire or statistically analysed user-acceptance experiment was identified in the available research evidence. Therefore, this thesis does not claim that the system has been formally validated for usability or operational effectiveness by DAPH officials or practising veterinarians.

Validation at this stage is better described as **functional and role-based validation**. Automated tests and operational workflows were used to confirm that users could access the functions intended for their roles, that restricted functions were protected,

and that disease-risk information could move through the required system workflows.

This distinction is important because successful implementation shows that the product can support the intended workflow, but formal field evaluation would still be required before claiming that the system improves real veterinary decision-making.

6.4 Integration results

Integration testing showed that the disease-risk component could operate within the wider R26-SE-013 application while retaining its own model and service boundaries.

The forecasting functionality was connected to shared backend infrastructure, authentication, persistence and frontend navigation. Cross-role workflows were also exercised through the application rather than testing the machine-learning service entirely in isolation.

The integration evidence recorded successful live HTTP execution of cross-role workflows and successful FMD forecast-record generation through the frozen prediction pipeline. Authentication with the integrated MongoDB-backed environment was also demonstrated for DAPH and Veterinary Officer roles at the documented integration checkpoint.

An important integration principle was that the forecasting models and scientific artefacts were not retrained or modified as part of system integration. The existing model files, scalers, feature order, thresholds and prediction rules remained separated from application-level workflow changes. This reduced the risk that software integration would unintentionally alter previously evaluated model behaviour.

The component also exposes an outbreak-status interface that can provide disease, district and period information to related components without allowing those components to directly depend on the internal forecasting model implementation.

Some integration work remained operational rather than scientific. The handover evidence identified human browser acceptance and additional runtime smoke validation as activities that still required completion at that checkpoint. These

remaining tasks do not alter the research model results, but they are relevant when considering the maturity of the complete application.

6.5 Summary of findings

The results show different levels of success across the two diseases.

For FMD, the strongest result was obtained from the 31-feature Logistic Regression variant containing recent district-level outbreak history. It achieved a three-fold walk-forward mean ROC-AUC of **0.8120** and a held-out 2024 ROC-AUC of **0.8166**, with **67.9% recall** at the operational threshold of 0.40.

For LSD, the calibrated 28-feature ElasticNet model produced more moderate performance, with a five-fold LOYO mean ROC-AUC of **0.6136** and PR-AUC of **0.2699**. The result supports the presence of predictive information in the combined feature set but also reflects the limitations of a shorter and more imbalanced surveillance history.

The research also demonstrated that model outputs could be explained using direct log-odds contributions and that Mondrian conformal prediction provided useful offline uncertainty coverage for FMD. At the product level, both disease models were integrated into API-based prediction services and role-oriented interfaces, with automated testing supporting the technical reliability of the implementation.

Overall, the findings support the component as an **advisory district-level disease-risk system**. They do not support interpreting the outputs as confirmed diagnosis or as validated multi-month future disease forecasts. The strongest contribution lies in combining sparse surveillance data, climate and spatial context, time-aware evaluation, interpretable model outputs and an operational software implementation within a single component.

7 DISCUSSION

7.1 Interpretation of findings

The findings show that district-level livestock disease risk can be estimated from a combination of surveillance history, climatic conditions, seasonal information and spatial context, but the strength of this relationship differs between FMD and LSD.

The FMD results were more encouraging. The 30-feature Logistic Regression model provided a useful production baseline, while the 31-feature variant improved discrimination after own_outbreak_lag1 was added. The increase in walk-forward ROC-AUC from 0.7833 to 0.8120 indicates that recent outbreak activity within the same district contains useful predictive information beyond climate and neighbouring-district information alone. The held-out 2024 ROC-AUC of 0.8166 further supports the presence of a meaningful predictive signal in the available FMD data.

This improvement should still be interpreted carefully. own_outbreak_lag1 represents recent disease persistence, not a causal mechanism. An outbreak in one month may increase the probability of another recorded outbreak in the next month because disease activity can continue across reporting periods. The model therefore benefits from temporal dependence in the surveillance data, but this does not mean that previous outbreak status alone explains future disease occurrence.

The threshold analysis also showed that model usefulness depends on how the prediction is operationalised. At a threshold of 0.40, the enhanced FMD model identified 67.9% of the observed outbreaks in the 2024 fold. Lowering the threshold improved outbreak detection but also increased the number of false-positive alerts. In a veterinary surveillance setting, this trade-off can be acceptable because a false alert can be investigated, whereas a missed outbreak may delay attention. The threshold should therefore be understood as an operational decision rather than a fixed characteristic of the model.

The LSD findings were more limited. The calibrated 28-feature ElasticNet model achieved a mean LOYO ROC-AUC of 0.6136 and PR-AUC of 0.2699. These results

indicate that useful predictive information exists in the feature set, but the performance is not strong enough to support claims of highly reliable prediction. The shorter study period and smaller number of positive LSD observations make the model more sensitive to differences between individual years.

The comparison between FMD and LSD also suggests that the same modelling strategy should not be assumed to work equally well for both diseases. FMD benefited more clearly from recent outbreak history and the longer observation period. LSD had fewer years of data and more limited positive observations, which reduced the amount of information available for training and evaluation.

The uncertainty analysis provided another useful finding. Offline Mondrian conformal evaluation for FMD achieved approximately 94.9% empirical coverage for outbreak cases at the selected nominal level. This suggests that class-conditional conformal prediction has value for future uncertainty support. However, because the current production system does not execute the full conformal procedure during runtime, the result is best interpreted as methodological evidence rather than as a deployed product capability.

7.2 Comparison with literature baselines and existing solutions

The results of this study are consistent with earlier livestock-disease research showing that disease occurrence is influenced by a combination of temporal, environmental, spatial and host-related factors.

For FMD, previous studies have shown that disease risk is spatially structured and that livestock density, movement, seasonal conditions and environmental factors can influence outbreak patterns [4], [5]. The present results support this general view. The FMD model performed better when spatial outbreak history and recent district-level outbreak persistence were included alongside climatic variables. This suggests that environmental conditions alone are not sufficient to represent the short-term structure of reported FMD activity.

For LSD, earlier studies have reported useful predictive performance from meteorological, livestock-density and geospatial variables [6]–[9]. The present research also found value in combining these categories of information, particularly after adding spatial outbreak variables and recent outbreak history. However, the lower LOYO performance obtained in this study highlights an important difference between a strong result under one validation design and reliable performance across separate years.

This is one area where the present study takes a more cautious position than some previous machine-learning work. Rather than relying only on a random split or full-dataset evaluation, performance was examined using walk-forward evaluation for FMD and year-separated LOYO evaluation for LSD. The resulting scores were lower than some values reported in related literature, but they provide a more realistic indication of how the models behave when evaluated outside the data used for fitting.

The study also compared the FMD classifier with simple baselines. The machine-learning model outperformed majority-class, lag-one persistence and seasonal-naive strategies, indicating that the combined feature set captured information beyond simple repetition of previous disease status or calendar seasonality.

Another difference concerns interpretability. Many studies use complex ensemble or black-box models to maximise predictive accuracy. In this research, Logistic Regression and ElasticNet were retained because they offered acceptable predictive performance together with direct interpretability. This aligns with the argument that interpretable models are preferable in decision-support settings when they can provide adequate performance [18].

The final system therefore differs from conventional surveillance tools in two ways. First, it combines multiple contextual variables rather than relying only on reported case counts. Second, it converts the resulting prediction into an operational district-level risk output that can be presented through a role-based software interface. However, unlike a true long-horizon forecasting system, the implemented future-date functionality relies on historical same-calendar-month proxies rather than future meteorological forecasts.

7.3 Research contribution

The main research contribution is the development and evaluation of a district-month livestock disease-risk framework for FMD and LSD using Sri Lankan surveillance and environmental information.

The first contribution is the reconstruction of sparse surveillance data into complete analytical district-month grids. This made it possible to combine veterinary records with climate, seasonal, livestock and spatial variables in a consistent format.

The second contribution is the integration of multiple forms of context within one modelling pipeline. The final feature representation included concurrent and lagged climate variables, monsoon and seasonal indicators, livestock density, large-scale climate indices, neighbouring outbreak activity and recent district-level disease history.

A third contribution is the evaluation strategy. The study did not rely only on random partitioning or full-dataset results. FMD was assessed using expanding-window walk-forward evaluation, while LSD was assessed using year-separated LOYO validation. This exposed differences between models that would have been less visible under a simpler evaluation design.

The analysis of own_outbreak_lag1 also provided a specific empirical contribution. For FMD, the addition of this single target-history feature increased the mean walk-forward ROC-AUC from 0.7833 to 0.8120. This suggests that recent outbreak persistence is an important part of short-term district-level risk estimation.

The study also contributed uncertainty and interpretation methods. Mondrian conformal prediction was evaluated offline for FMD, while exact log-odds attribution was used for runtime explanation. This combination provides a clearer connection between the statistical model and the risk information shown to users.

Finally, the research process itself involved identifying and removing methods that could not be supported adequately. Earlier heuristic SIR-derived features were removed, and unsupported in-sample claims were not retained in the final evaluation. This strengthened the traceability between the reported results and the implemented models.

7.4 Product contribution

The product contribution lies in converting the research pipeline into a usable software component within the wider R26-SE-013 platform.

The trained FMD and LSD models were integrated into FastAPI services rather than remaining as standalone notebooks or experimental scripts. The backend supports disease-specific prediction routes, district-wide forecast routes, model selection, fallback behaviour, severity evaluation and explanation generation.

The frontend presents the results through role-oriented interfaces. Veterinary users can access district-focused risk and follow-up information, while DAPH-oriented views provide broader surveillance summaries, data-quality information and national-level oversight. The software therefore translates model output into a form that is more appropriate for operational decision support than a raw probability alone.

Another product contribution is the separation between scientific model logic and application workflow. Model files, scalers, feature definitions and thresholds remain within the forecasting component, while the wider platform interacts with them through defined API interfaces. This reduces the risk that integration work changes the scientific behaviour of the models.

The system also records the provenance of prediction inputs. A result based on an exact district-month record can be distinguished from one generated using historical proxy data. This is particularly important for future-date risk projections, where the software uses the latest available same-calendar-month observations rather than actual future weather data.

The product should still be described as a research decision-support system rather than a fully operational national surveillance platform. The available evidence supports software integration, role-based workflows and automated testing, but formal field deployment and institutional validation remain outside the completed scope.

7.5 Limitations and threats to validity

Several limitations affect how the results should be interpreted.

The first limitation is the amount and balance of the surveillance data. Outbreak-positive district-months form a relatively small proportion of both datasets. This is especially important for LSD, where only five years of data were available. A small number of positive observations increases uncertainty in both model training and evaluation.

A second limitation concerns temporal generalisation. Although FMD showed relatively strong walk-forward performance, the LSD results were more variable across held-out years. This means that average performance should not be interpreted as evidence that the model will perform equally well under all future epidemiological conditions.

A third limitation is the use of recent outbreak-history variables. Features such as own_outbreak_lag1 and neighbouring outbreak lags improve predictive performance, but they introduce dependence on timely surveillance reporting. If previous-month disease information is missing, delayed or inaccurate, the enhanced model may not be usable. The production system partly addresses this by falling back to models that do not require the missing target-history feature.

Another limitation concerns the future forecast endpoints. Several active model features use climatic information from the target month. When future observed values are unavailable, the software substitutes historical same-calendar-month conditions. These results are therefore climatological risk projections rather than genuine weather-informed future forecasts.

Data provenance is also a concern. Formal independent certification of some DAPH-derived records was not available during the study. The datasets were sufficient for model development and evaluation, but the absence of complete provenance documentation limits the strength of claims that can be made about institutional data validation.

The fallback imputation process also represents a possible source of bias. Historical medians and same-month proxy records are practical for software operation, but they do not fully reproduce the information conditions that would exist in a strictly prospective system.

External validity remains another limitation. The models were developed and evaluated using Sri Lankan district-level data. They were not tested on an independent external-country dataset, so their performance should not be assumed to transfer directly to other geographical settings. External spatial validation was identified as a remaining evidence gap.

Finally, formal veterinary user validation was not completed. The application includes role-based workflows for veterinary and DAPH users, but the evidence package identifies expert usability review as pending. Consequently, the study demonstrates technical and functional validity rather than proven real-world improvement in veterinary decision-making.

7.6 Implications

The findings have both research and practical implications.

From a research perspective, the results show that climate information alone is not sufficient for short-term livestock disease-risk estimation. Spatial context and recent outbreak history contributed substantially to model performance, particularly for FMD. Future work on similar surveillance systems should therefore consider combining environmental predictors with epidemiological persistence and spatial relationships rather than treating each district-month as an independent observation.

The results also reinforce the importance of time-aware evaluation. Strong performance obtained from random or in-sample testing may not reflect how a model behaves in later or previously unseen periods. The weaker and more variable LSD results illustrate why year-separated validation should be retained even when it produces less impressive scores.

From a practical perspective, the FMD model may be useful for prioritising districts for closer veterinary attention, especially when combined with professional judgement and current field information. The system could help identify areas where surveillance, investigation or preventive planning deserves additional attention. It should not be used as a substitute for clinical confirmation or official disease-control procedures.

The LSD results suggest a more cautious role. The model may support broad risk awareness, but its current evidence is not strong enough to justify reliance on it as a stand-alone early-warning tool.

The software implementation also demonstrates how predictive research can be integrated into a wider livestock-health platform without exposing the model as an unexplained black box. Role-based interfaces, data-quality information and direct feature explanations provide a more responsible way of presenting statistical risk information.

Overall, the practical implication is not that the system can predict livestock outbreaks with certainty. Its value lies in organising multiple sources of historical and environmental information into a structured district-level risk signal that can support, rather than replace, veterinary surveillance and professional judgement.

8 CONCLUSION AND FUTURE DIRECTIONS

8.1 Conclusion

This research developed and evaluated a climate-informed, district-level disease-risk component for Foot-and-Mouth Disease (FMD) and Lumpy Skin Disease (LSD) in Sri Lanka. The work addressed the difficulty of extracting useful risk information from sparse and imbalanced veterinary surveillance records by combining disease history with climatic, seasonal, livestock-related and spatial information.

The research showed that the two diseases could not be treated as identical modelling problems. FMD produced the stronger results. The 30-feature Logistic Regression model provided a stable production baseline, while the addition of recent district-level outbreak history through own_outbreak_lag1 produced an enhanced 31-feature model. This variant achieved a mean walk-forward ROC-AUC of 0.8120 and a held-out 2024 ROC-AUC of 0.8166. At the selected threshold of 0.40, it identified 67.9% of the observed FMD outbreak district-months.

For LSD, the final 28-feature Platt-calibrated ElasticNet model achieved a five-fold Leave-One-Year-Out mean ROC-AUC of 0.6136 and PR-AUC of 0.2699. Although this was an improvement over the simpler climate-oriented baseline, the results were more moderate than those obtained for FMD. The limited observation period and small number of positive LSD records therefore require the model to be interpreted more cautiously.

The study also showed the importance of looking beyond a single prediction score. Threshold analysis was used to examine the trade-off between missed outbreaks and false alerts, while direct linear log-odds decomposition allowed individual predictions to be explained using the fitted model itself. Mondrian conformal prediction was evaluated offline for FMD as an uncertainty-quantification method, although the complete conformal procedure is not currently deployed as part of live inference.

At product level, the trained models were incorporated into the wider R26-SE-013 platform through FastAPI services and role-oriented frontend interfaces. The

resulting system can generate district-level disease-risk estimates and present them as advisory information to support veterinary monitoring and prioritisation.

The completed component should therefore be viewed as a **veterinary decision-support system rather than an autonomous disease-control system**. Its main value lies in bringing together previously separated sources of surveillance, environmental and historical information into an interpretable district-level risk estimate. It does not replace clinical diagnosis, field investigation or decisions made by veterinary authorities.

8.2 Achievement of objectives

The objectives established at the beginning of the study were addressed to different degrees during the research and implementation process.

Table 7 : Achievement of Objectives

Objective	**Achievement**
Reconstruct FMD and LSD surveillance records into district-month datasets	**Achieved.** Complete analytical datasets were prepared for all 25 districts, covering 2017–2024 for FMD and 2020–2024 for LSD.
Integrate climate, seasonality, livestock, teleconnection, spatial and historical outbreak information	**Achieved.** These information sources were incorporated into the model feature pipelines.
Develop and evaluate machine-learning models for imbalanced outbreak-risk classification	**Achieved.** Logistic Regression-based FMD models and ElasticNet-based LSD models were developed and evaluated.
Examine the effect of recent district-level outbreak autocorrelation	**Achieved.** Adding `own_outbreak_lag1` improved FMD mean walk-forward ROC-AUC from 0.7833 to 0.8120.
Evaluate models using time-aware and year-separated procedures	**Achieved.** FMD used expanding-window walk-forward evaluation, while LSD

	used Leave-One-Year-Out evaluation.
Investigate an operational decision threshold	**Achieved.** A threshold of 0.40 was evaluated and retained for the implemented decision workflow.
Evaluate uncertainty and provide interpretable model outputs	**Partially achieved.** Mondrian conformal prediction was successfully evaluated offline for FMD, while runtime interpretability was implemented through exact log-odds decomposition. Full conformal uncertainty remains outside the current live inference pipeline.
Integrate the models into the wider R26-SE-013 platform	**Achieved.** The component was exposed through backend services and integrated with the frontend decision-support workflow.

The objectives were therefore largely achieved, while uncertainty deployment and broader operational validation remain areas for further work.

8.3 Individual research and product contributions

The individual research contribution covers the complete development path of the climate-informed disease-risk component.

From a research perspective, the work included reconstruction of the surveillance datasets, integration of environmental and epidemiological data, development of the spatio-temporal feature pipeline, model comparison, threshold analysis, evaluation of target autocorrelation, year-aware model validation, offline uncertainty analysis and interpretable prediction design.

An important contribution was the separation of stronger evidence from weaker experimental findings. Earlier approaches that were not sufficiently supported, including heuristic SIR-derived features, were not retained as final research claims. Likewise, the final thesis distinguishes between a model that performs concurrent district-month nowcasting and the future-facing software function that uses historical same-calendar-month conditions as a climatological proxy.

From a product perspective, the work converted the research models into a reusable software component rather than leaving them as isolated experiments. The implementation includes disease-specific FastAPI prediction services, model and feature routing, threshold-based Stage 2 execution, data-quality handling and exact model explanations.

The component was also integrated into the wider application using role-oriented interfaces and shared system infrastructure. This provides a practical connection between the research outcome and the broader objective of supporting livestock-health monitoring through a centralized digital platform.

8.4 Future directions

Several improvements would strengthen both the scientific quality and practical usefulness of the component.

The priority is to extend the surveillance datasets with additional years of verified FMD and LSD records. This is particularly important for LSD, where only five years of data were available. Additional outbreak-positive observations would allow more reliable temporal evaluation and may improve the stability of the classifier.

A second direction is to move from climatological future projections toward genuine future disease forecasting. The current future-date endpoints use historical same-calendar-month environmental conditions when target-period observations are unavailable. A future version could integrate reliable weather forecasts or seasonal climate predictions so that future disease risk is estimated using information genuinely available before the target period.

The uncertainty layer could also be extended. Mondrian conformal prediction produced encouraging FMD coverage during offline evaluation, but a production implementation would require a carefully maintained calibration set, monitoring under distribution shift and sufficient positive outbreak examples before uncertainty sets could be presented reliably to users.

Further work should also examine richer representations of spatial disease movement. The current neighbouring-district features are based mainly on administrative adjacency. Future models could incorporate livestock transport routes, markets, road connectivity or animal-movement information to represent transmission pathways more directly.

Formal external validation is another priority. The current models were developed and evaluated using Sri Lankan district-level data. Testing on additional surveillance periods or an independent external dataset would provide stronger evidence about generalisability.

Stakeholder validation should also be completed. Veterinary officers and DAPH officials could evaluate the usefulness, clarity and practical suitability of the risk displays through structured usability studies or expert review. The existing evidence package identifies veterinary expert evaluation as an area that remains incomplete.

Finally, future development should strengthen data provenance and operational data pipelines. Formal documentation of surveillance-data acquisition, periodic data updates, model-drift monitoring and secure production deployment would be necessary before considering the component for wider institutional use.

These directions would allow the current research prototype to progress from a retrospective, climate-informed district-level risk system toward a more continuously updated and prospectively validated livestock disease early-warning platform.

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GLOSSARY

Term	Definition
Climatological Risk Projection	A future-period outbreak-risk estimate produced using historical environmental conditions from the same calendar month when actual future climatic observations are unavailable.
Concurrent Nowcasting	Prediction of outbreak probability for a district-month using climatic information from that same target month together with recent historical and epidemiological variables.
Data Leakage	The unintended use of information during model development or evaluation that would not legitimately be available when predicting unseen data.
Decision Threshold	The probability cut-off used to convert a continuous outbreak probability into an operational outbreak-risk classification.
District-Month	The main analytical unit used in this research represents one administrative district during one calendar month.
Fallback Model	An alternative trained model used when information required by the preferred model, such as verified previous-month outbreak status, is unavailable.
Historical Same-Month Proxy	A historical observation from the same calendar month used to represent unavailable environmental conditions for a requested future period.
Log-Odds Decomposition	A direct interpretation method for a linear classifier in which the model decision score is separated into contributions from individual input features.
Mondrian Conformal Prediction	A class-conditional conformal prediction method in which uncertainty calibration is performed separately for different outcome classes.
Spatial Lag	A feature representing disease activity from geographically neighboring districts during an earlier period.
Stage 1 Classification	The first modelling stage that predicts the probability of outbreak occurrence for a district-month.
Stage 2 Classification	A conditional secondary modelling stage evaluated after the Stage 1 outbreak probability reaches the operational threshold.
Target Autocorrelation	Predictive dependence between current disease occurrence and earlier observations of the same disease, represented in this study through features such as own_outbreak_lag1.
Temporal Lag	A feature representing the value of a variable during one or more earlier time periods.

APPENDICES

Appendix A Supplementary Model and Technical Evidence

A.1 Final Model Configurations

The final disease-risk component used separate model configurations for FMD and LSD. The main production and fallback configurations are summarised below.

Disease / Stage	Model	Feature Configuration	Operational Role
FMD Stage 1	Logistic Regression	30 features	Default production baseline
FMD Stage 1 Enhanced	Logistic Regression	31 features including `own_outbreak_lag1`	Used when verified previous-month outbreak information is available
FMD Stage 2	Random Forest Classifier	Severity feature set	Conditional LOW, MEDIUM or HIGH advisory severity
LSD Stage 1	Platt-calibrated ElasticNet Logistic Regression	28 features	Primary LSD outbreak-probability model
LSD Stage 1 Fallback	ElasticNet Logistic Regression	27 features	Used when previous-month outbreak status is unavailable
LSD Stage 2	Logistic Regression	Severity/suppression configuration	Used mainly as a quiet-period false-alarm suppressor

The Stage 1 models estimate the probability of outbreak occurrence for a requested district-month. An operational threshold of 0.40 is applied before Stage 2 is

conditionally evaluated. The LSD Stage 2 output is interpreted cautiously because it has not demonstrated reliable severity discrimination during active outbreak periods.

A.2 FMD Held Out 2024 Classification Results

30-Feature Baseline at Threshold 0.40

Actually / Predicted	No Outbreak	Outbreak
No Outbreak	172	50
Outbreak	27	51

- Recall: 65.4%
- Precision: 50.5%
- F1-score: 0.570
- ROC-AUC: 0.7570
- PR-AUC: 0.5461

31-Feature Autocorrelation Variant at Threshold 0.40

Actually / Predicted	No Outbreak	Outbreak
No Outbreak	175	47
Outbreak	25	53

- Recall: 67.9%
- Precision: 53.0%
- F1-score: 0.596
- ROC-AUC: 0.8166
- PR-AUC: 0.6968

The enhanced 31-feature model improved both discrimination and threshold-specific outbreak detection compared with the 30-feature baseline. These detailed values supplement the results presented in Chapter 6.

A.3 Main Prediction and Integration Endpoints

Endpoint	Purpose
`POST /api/v1/risk-forecasting/predict/fmd`	Predict FMD outbreak probability for an individual district-month
`POST /api/v1/risk-forecasting/predict/lsd`	Predict LSD outbreak probability for an individual district-month
`POST /api/v1/risk-forecasting/forecast/fmd`	Generate district-wide FMD climatological risk projections
`POST /api/v1/risk-forecasting/forecast/lsd`	Generate district-wide LSD climatological risk projections
`GET /outbreak-status/{disease}/{district}/{year}/{month}`	Provide outbreak-status information for cross-component integration

The /forecast endpoints should not be interpreted as independently validated future-weather forecasting. For future requested periods, the implementation uses historical same-calendar-month environmental conditions as climatological proxies when actual target-period observations are unavailable.

A.4 Automated Test Summary

Test Area	Tests Passed
Backend	371
Frontend	491
Total	**862**

The final recorded automated test execution contained 862 passing tests with no recorded failures. The backend tests covered the forecasting component together with

shared authentication, routes, database behavior and operational workflows, while frontend tests covered role-oriented views, navigation and application behavior.

Appendix B Academic integrity and AI use disclosure

Generative AI and automated software-assistance tools were used during this research as supporting tools for language refinement, document structuring, technical consistency checking, code inspection, and identification of relevant implementation evidence. These tools were not treated as authoritative sources for research findings.

Model metrics, dataset properties, implementation details, API behaviour, and software-testing results reported in this thesis were checked against project source code, trained model artefacts, datasets, evaluation scripts, testing evidence, or other available project records. Academic references were independently checked against peer-reviewed publications or authoritative institutional sources before inclusion.

AI-assisted outputs were reviewed, revised, and incorporated by the author where appropriate. The author remains responsible for the research design, software development, analysis, interpretation of findings, citations, written content, and conclusions presented in this thesis.

Tool	Purpose	Verification Performed
ChatGPT	Thesis structuring, language refinement, consistency checking, clarification of technical concepts, and support in organizing evidence	Content was reviewed against project files, source code, research results, and verified academic references
Coding / AI Assistant	Codebase inspection and extraction of implementation details where applicable	Claims were checked against source files, model artefacts, routes, evaluation scripts, and automated test evidence

Representative AI-assisted workflows included reviewing chapter structure, checking consistency between model results and implementation descriptions, refining academic wording, and verifying whether reported software behavior matched the final implementation. AI-generated text was not accepted without review.